



On LASSO for predictive regression

Ji Hyung Lee^a, Zhentao Shi^{b,*}, Zhan Gao^c

^a Department of Economics, University of Illinois, 1407 W. Gregory Dr., 214 David Kinley Hall, Urbana, IL 61801, USA

^b Department of Economics, The Chinese University of Hong Kong, Sha Tin, New Territories, Hong Kong Special Administrative Region of China

^c Department of Economics, University of Southern California, Kaprielian Hall, 3620 South Vermont Avenue, Los Angeles, CA 90089, USA

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ABSTRACT

Explanatory variables in a predictive regression typically exhibit low signal strength and various degrees of persistence. Variable selection in such a context is of great importance. In this paper, we explore the pitfalls and possibilities of the LASSO methods in this predictive regression framework. In the presence of stationary, local unit root, and cointegrated predictors, we show that the adaptive LASSO cannot asymptotically eliminate all cointegrating variables with zero regression coefficients. This new finding motivates a novel post-selection adaptive LASSO, which we call the *twin adaptive LASSO* (TAlasso), to restore variable selection consistency. Accommodating the system of heterogeneous regressors, TAlasso achieves the well-known oracle property. In contrast, conventional LASSO fails to attain coefficient estimation consistency and variable screening in all components simultaneously. We apply these LASSO methods to evaluate the short- and long-horizon predictability of S&P 500 excess returns.

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1. Introduction

Predictive regressions are used extensively in empirical finance. A leading example is the stock return regression, for which predictability has long been a primary goal. We focus on this type of predictive regression in this paper. The first central econometric issue in these models is severe test size distortion in the presence of highly persistent predictors coupled with regression endogeneity. When persistence and endogeneity are high, the conventional inferential apparatus designed for stationary data can be misleading. Another major challenge in predictive regressions is the low signal-to-noise ratio (SNR). One explanation is that the noise in short-term predictive models may obscure the signal given by slow-moving (persistent) predictors.

Even though certain predictors have been suggested based on economic theory, there is no consensus regarding the predictive ability of any predictor. A researcher may use *ex ante* a pool of candidate regressors, hoping to identify a few important regressors. However, the more variables the researcher includes, the greater is the need for a data-driven routine for variable selection as many of these variables *ex post* demonstrate little or no predictive ability because of the competitive nature of the market. A low SNR naturally calls for variable selection.

Advancements in machine learning techniques, driven by an unprecedented abundance of data sources from many disciplines, offer valuable opportunities for economic data analysis. In the era of big data, shrinkage methods are becoming

* Corresponding author.

E-mail addresses: jihyung@illinois.edu (J. Lee), zhentao.shi@cuhk.edu.hk (Z. Shi), zhangao@usc.edu (Z. Gao).

increasingly popular in econometric inference and prediction because of their variable selection and regularization properties. In particular, the *least absolute shrinkage and selection operator* (LASSO; Tibshirani, 1996) has received intensive study in the past two decades.

This paper investigates LASSO methods in predictive regressions. The LASSO methods are attractive in predictive regressions because they enable researchers to identify pertinent and exclude irrelevant predictors. However, time series predictors in predictive regressions carry heterogeneous degrees of persistence, which we call *mixed roots*. Some may exhibit short memory (e.g., Treasury bills), whereas others may be highly persistent (e.g., most financial/macro predictors). Moreover, a multitude of persistent predictors can be cointegrated. For example, the dividend price (DP) ratio is essentially a cointegrating residual between dividend and price, and the so-called *cay data* (Lettau and Ludvigson, 2001) is another cointegrating residual between consumption, asset holdings, and labor income.

Given the difficulty of classifying time series predictors, we take an agnostic view of the types of the regressors and examine whether the LASSO methods can cope with heterogeneous regressors. We consider a predictive regression with three categories of predictors: short memory ($I(0)$) regressors, cointegrated local-to-unity persistent regressors, and non-cointegrated persistent regressors. Our focus is on variable selection. It is well known that in cross-sectional regressions, the *adaptive LASSO* (Alasso; Zou, 2006) enjoys *variable selection consistency*, which means that the estimated zeros exactly coincide with the true zero coefficients when the sample size is sufficiently large. The heterogeneous time series regressors challenge the conventional wisdom.

The main contribution of this paper is its study and refinement of Alasso in the time series context. As we consider a fixed number of regressors, we use OLS as Alasso's initial estimator. Alasso, with a proper choice of tuning parameter, consistently selects the non-cointegrating stationary and persistent variables, but it may over-select inactive cointegrating variables, where “inactive” means that these variables have zero regression coefficients, so they are excluded in the true data-generating process (DGP). The convergence rate of the OLS estimator is $\sqrt{n}$ for the cointegrated variables, and the resulting Alasso penalty weight is too small to eliminate the true inactive cointegrated groups because each cointegrating time series behaves as a local unit root process. Nevertheless, as is formally stated in Theorem 3, with probability approaching one (wpa1), variables forming an inactive cointegrating group cannot all survive Alasso's selection.

This finding suggests a straightforward method of reclaiming the desirable oracle property: simply run a second-round Alasso on the variables selected by the first-round Alasso. As the first-round Alasso has broken the cointegration relationship between inactive cointegrating variables, the over-selected variables become non-cointegrated local-to-unity regressors in the second-round regression. In the post-selection OLS, the speed of convergence of these variables is boosted from the slow $\sqrt{n}$ rate to the fast n rate. Then, the second-round Alasso can successfully suppress all of the inactive coefficients wpa1. We call this post-selection Alasso procedure *twin adaptive LASSO* and abbreviate it as TALasso.

TAlasso achieves the oracle property (Fan and Li, 2001), which implies that the rate of convergence is the same as the OLS when the relevant variables are known in advance and that the variable selection is consistent in the presence of a mixture of heterogeneous regressors. To the best of our knowledge, this paper is the first to establish these desirable properties in a nonstationary time series context. Note that the name TALasso distinguishes it from the *post-selection double LASSO* (Belloni et al., 2014). Double LASSO is named for the double inclusion of selected variables in cross-sectional data to correct for LASSO's shrinkage bias in order to achieve uniform statistical inference. In contrast, our TALasso features double exclusion in predictive regressions with mixed roots.

In developing asymptotic theory, we consider a linear process of time series innovations that encompasses a general ARMA structure arising in many practical applications, such as long-horizon return prediction. To focus on the distinctive feature of nonstationary time series, we adopt a simple asymptotic framework in which the number of regressors p is fixed and the number of time periods n passes to infinity. Our exploration in this paper is a stepping stone toward automated variable selection in high-dimensional predictive regressions.

We also explore the *plain LASSO* (Plasso henceforth; Tibshirani, 1996), the *standardized LASSO* (Slasso, in which the l_1 -penalty is multiplied by the sample standard deviation of each regressor) in the predictive regression with heterogeneous regressors. We call the phenomenon *variable screening effect* whereby an estimator coerces some coefficients to exactly zero (Fan and Lv, 2008). We find that neither Plasso nor Slasso maintains variable screening and consistent coefficient estimation in all of the components simultaneously, as Plasso imposes equal penalty weight regardless of the nature of the regressor, whereas Slasso's penalty only changes according to the scale variation of individual regressors and ignores their connection via cointegration.

In Monte Carlo simulations, we use various DGPs to examine the finite sample performance of Alasso/TAlasso compared with Plasso/Slasso, assessing their mean squared prediction errors and variable selection success rates. TALasso is much more capable in selecting the correct model, thus improving prediction accuracy. These LASSO methods are further evaluated in a real data application. We use the 12 predictors from Welch and Goyal (2008) to predict the S&P 500 returns. These 12 predictors exemplify the necessity of considering the three types of heterogeneous regressors. Please see Section 6 for the details. Alasso and TALasso attain better forecasting performance than their competitors and are robust in various estimation windows and prediction horizons.

Literature review. Since the original LASSO paper by Tibshirani (1996) and the basis pursuit of Chen et al. (2001), a variety of important extensions of LASSO have been proposed, such as Alasso (Zou, 2006) and the elastic net (Zou and Hastie, 2005). In econometrics, Caner (2009), Caner and Zhang (2014), Shi (2016), Su et al. (2016), and Kock and Tang (2019) use LASSO-type procedures in cross-sectional and panel data models. Belloni and Chernozhukov (2011), Belloni et al.

(2012), Belloni et al. (2014), and Belloni et al. (2015) develop methodologies and inferential theories in a variety of microeconomic settings.

Ng (2013) surveys the variable selection methods in predictive regressions. Unlike the extensive literature on cross-sectional environments, the theoretical properties of shrinkage methods have received relatively little attention in time series models despite great empirical interest in macro and financial applications (Giannone et al., 2018; Gu et al., 2020; Chincio et al., 2019; Rapach et al., 2019; Freyberger et al., 2020). Medeiros and Mendes (2016) study Alasso in high-dimensional stationary time series, and Kock and Callot (2015) discuss LASSO in a vector autoregression (VAR) system. In time series forecasting, Inoue and Kilian (2008) apply various model selection and model averaging methods to forecast U.S. consumer price inflation. Sun et al. (2020) study model averaging methods under structural changes, and Hirano and Wright (2017) develop a local asymptotic framework with independently and identically distributed (iid) orthonormalized predictors to study the risk properties of several machine learning estimators. There are even fewer papers on LASSO with nonstationary data. Caner and Knight (2013) discuss the bridge estimator, a generalization of LASSO, for the augmented Dickey–Fuller test in autoregressions, and Kock (2016) studies Alasso under the same setup. Earlier works on variable selection in the $I(1)$ case utilize the different stochastic limit of penalty functions with stochastically trending regressors, such as Phillips and Ploberger (1994) and Ng and Perron (2001). For example, the stochastic limit of penalty factors is different with integrated regressors compared with the stationary case (Ng and Perron, 2001, pp. 1529–1530).

In predictive regressions, Kostakis et al. (2014), Lee (2016), and Phillips and Lee (2013, 2016) provide inferential procedures in the presence of multiple predictors with various degrees of persistence. Xu (2018) studies variable selection and inference with possible cointegration between the $I(1)$ predictors. In a vector error correction model (VECM), Liao and Phillips (2015) use Alasso for cointegration rank selection. Under a similar setting with high-dimensional $I(0)$ and $I(1)$ regressors and one cointegration group, Koo et al. (2020) investigate Plasso's variable estimation consistency and non-standard asymptotic distribution. We differ from Koo et al. (2020) in two aspects. (i) In an asymptotic framework that accommodates high-dimensional regressors, they focus on the rate of convergence of Plasso. In contrast, our focus is the oracle property in the mixed root model with three types of regressors. (ii) The unit root variables enter their model via the cointegrating groups only, so they do not deal with the balance of the predictive regression, whereas in our analysis, we allow local-to-unity persistent variables and local-to-zero small coefficients. Finally, Smeekes and Wijler (2018) demonstrate LASSO's numerical performance via simulations and empirical examples, while we are the first to systematically explore the theory concerning variable selection under mixed regressor persistence.

Notation. We use standard notation. We define $\|\cdot\|_1$ and $\|\cdot\|$ as the usual vector l_1 - and l_2 -norms, respectively. The arrows $\Rightarrow$ and $\xrightarrow{p}$ represent weak convergence and convergence in probability, respectively. $\asymp$ indicates “of the same asymptotic order,” and $\sim$ signifies “being distributed as” either exactly or asymptotically, depending on the context. $\lfloor \cdot \rfloor$ is the floor function. $a \wedge b = \min\{a, b\}$. The symbols $O(1)$ and $o(1)$ ($O_p(1)$ and $o_p(1)$) denote (stochastically) asymptotically bounded or negligible quantities. For a generic set M , let $|M|$ be its cardinality. For a generic vector $\theta = (\theta_j)_{j=1}^p$ with $p \geq |M|$, let $\theta_M = (\theta_j)_{j \in M}$ be the subvector of θ associated with the index set M . I_p is the $p \times p$ identity matrix, $\mathbf{1}_p$ is a column of p ones, and $I(\cdot)$ is the indicator function.

The rest of this paper is organized as follows. Section 2 introduces unit root regressors into a simple LASSO framework to fix ideas of Alasso. This model is substantially generalized in Section 3 to include $I(0)$, local-to-unity, and cointegrated regressors, and the asymptotic properties of Alasso and TAllasso are established. The mixed root model is carried over into Section 4 to explore the behaviors of Plasso and Slasso. The theoretical results are confirmed through a set of empirically motivated simulation designs in Section 5. Finally, we examine stock return predictions via these LASSO methods in Section 6.

2. Adaptive LASSO with unit roots

In this section, we study LASSO with p unit root regressors. To fix ideas, we investigate the asymptotic behavior of Alasso under a simple nonstationary regression model to better understand the technical issues arising from nonstationary predictors. Section 3 generalizes the model to include the $I(0)$, (local) unit roots, and cointegrated predictors.

2.1. Model

Assume that the dependent variable y_i is generated from a linear model

$$y_i = \sum_{j=1}^p x_{ij} \beta_{jn}^* + u_i = x_i \beta_n^* + u_i, \quad i = 1, \dots, n, \quad (1)$$

where n is the sample size. The $p \times 1$ true coefficient is $\beta_n^* = (\beta_{jn}^*)_{j=1}^p = \beta_j^{0*}/n^{\delta_j}$, where $\beta_j^{0*} \in \mathbb{R}$ is a fixed constant independent of the sample size and $\delta_j \in (0, 1)$ is the local-to-zero rate, following Phillips and Lee (2013) and Timmermann and Zhu (2017). If $\beta_j^{0*} = 0$, then β_{jn}^* remains zero regardless of the sample size; if $\beta_j^{0*} \neq 0$, then it varies with n . This type of local-to-zero coefficient is designed to balance the $I(0)$ – $I(1)$ relation between the stock return and unit root predictors,

and to model the weak SNR in predictive regressions (Phillips, 2015). The $1 \times p$ regressor vector $x_i = (x_{i1}, \dots, x_{ip})$ follows a pure unit root process

$$x_i = x_{(i-1)} + e_i = \sum_{k=1}^i e_k, \quad (2)$$

where $e_k = (e_{k1}, \dots, e_{kp})$ is the innovation. For simplicity, we assume the initial value $e_0 = 0$. We use the following iid assumption on the innovations.

Assumption 1. The innovations e_i and u_i are generated from

$$(e_i, u_i)' \sim iid(0, \Sigma),$$

where $\Sigma = \begin{pmatrix} \Sigma_{ee} & \Sigma_{eu} \\ \Sigma'_{eu} & \sigma_u^2 \end{pmatrix}$ is positive-definite.

The regression equation (1) can be equivalently written as

$$y = \sum_{j=1}^p x_j \beta_{jn}^* + u = X \beta_n^* + u, \quad (3)$$

where $y = (y_1, \dots, y_n)'$ is the $n \times 1$ response vector, $u = (u_1, \dots, u_n)'$, $x_j = (x_{1j}, \dots, x_{nj})'$, and $X = (x_1, \dots, x_p)$ is the $n \times p$ predictor matrix. This pure I(1) regressor model in (3) is a direct extension of the common predictive regression application with a single unit root predictor (e.g., DP ratio). The mixed roots case in Section 3 has multiple predictors and is more realistic in practice.

The literature focuses on the non-standard statistical inference caused by persistent regressors and weak signals. The asymptotic theory is usually confined to a small number of candidate predictors. Following the literature on predictive regressions, we consider the asymptotic framework in which p is fixed and the sample size $n \rightarrow \infty$. This simple asymptotic framework allows us to concentrate on the contrast between the standard iid setting and the predictive regression involving nonstationary regressors.

In this model, the unknown coefficients β_n^* can be obtained from the data by running OLS

$$\hat{\beta}^{ols} = \arg \min_{\beta} \|y - X\beta\|^2,$$

whose asymptotic behavior is well understood (Phillips, 1987). Assumption 1 implies the following functional central limit theorem:

$$\frac{1}{\sqrt{n}} \sum_{k=1}^{\lfloor nr \rfloor} \begin{pmatrix} e'_k \\ u_k \end{pmatrix} \Rightarrow \begin{pmatrix} B_e(r) \\ B_u(r) \end{pmatrix} \equiv BM(\Sigma). \quad (4)$$

To represent the asymptotic distribution of the OLS estimator, define $u_i^+ = u_i - \Sigma'_{eu} \Sigma_{ee}^{-1} e'_i$, and then $n^{-1/2} \sum_{i=1}^{\lfloor nr \rfloor} u_i^+ \Rightarrow B_{u^+}(r)$. By definition, $\text{cov}(e_{ij}, u_i^+) = 0$ for all j so that

$$\frac{X'u}{n} \Rightarrow \zeta := \int_0^1 B_e(r) dB_{u^+}(r) + \int_0^1 B_e(r) \Sigma'_{eu} \Sigma_{ee}^{-1} dB_e(r),$$

which is the sum of a (mixed) normal random vector and a non-standard random vector. The OLS limit distribution is

$$n(\hat{\beta}^{ols} - \beta_n^*) = \left(\frac{X'X}{n^2} \right)^{-1} \frac{X'u}{n} \Rightarrow \Omega^{-1} \zeta, \quad (5)$$

where $\Omega := \int_0^1 B_e(r) B_e(r)' dr$. This result implies that when we inflate $\hat{\beta}_j^{ols}$ by the factor n^{δ_j} so that its magnitude is comparable to the constant β_j^{0*} , we attain consistency in that

$$n^{\delta_j} (\hat{\beta}_j^{ols} - \beta_{jn}^*) = n^{\delta_j} \hat{\beta}_j^{ols} - \beta_j^{0*} = O_p(n^{\delta_j-1}) = o_p(1) \text{ for all } j \leq p.$$

Even if $\beta_j^{0*} \neq 0$, when δ_j is close to 1, the signal of x_j is weak, and therefore the rate of convergence is slow. However, some true coefficients β_j^{0*} in (3) could be exactly zero, where the associated predictors would be redundant (inactive) in the regression. Let $M^* = \{j : \beta_j^{0*} \neq 0\}$ be the index set of regressors relevant to the regression, $p^* = |M^*|$, and $M^{*c} = \{1, \dots, p\} \setminus M^*$ be the set of redundant regressors. For simplicity, we refer to M^* as the *active set*, meaning that it plays an active role in the regression, and we call M^{*c} the *inactive set*. If we had knowledge about M^* , ideally, we would estimate the unknown parameters by OLS in the active set M^* only. Define $\hat{\beta}^{ora} = (\hat{\beta}_{M^*}^{ora}, \hat{\beta}_{M^{*c}}^{ora})'$, where

$$\hat{\beta}_{M^*}^{ora} = \arg \min_{\beta} \|y - \sum_{j \in M^*} x_j \beta_j\|^2$$

and $\hat{\beta}_{M^{*c}}^{\text{ora}} = \mathbf{0}$. We call $\hat{\beta}^{\text{ora}}$ the *oracle estimator*, which is based on the infeasible “oracle” information of M^* . Eq. (5) implies that its asymptotic distribution is

$$n(\hat{\beta}^{\text{ora}} - \beta_n^*)_{M^*} \Rightarrow \Omega_{M^*}^{-1} \zeta_{M^*},$$

where Ω_{M^*} is the $p^* \times p^*$ submatrix $(\Omega_{jj'})_{j,j' \in M^*}$ and ζ_{M^*} is the $p^* \times 1$ subvector $(\zeta_j)_{j \in M^*}$.

2.2. Adaptive LASSO

We study the asymptotic behavior of Alasso in predictive regressions with these pure unit root regressors. Alasso for (1) is defined as

$$\hat{\beta}^A = \arg \min_{\beta} \left\{ \|y - X\beta\|^2 + \lambda_n \sum_{j=1}^p \hat{\tau}_j |\beta_j| \right\}, \quad (6)$$

where the weight $\hat{\tau}_j = |\hat{\beta}_j^{\text{init}}|^{-\gamma}$ for some initial estimator $\hat{\beta}_j^{\text{init}}$, and λ_n and γ are the two tuning parameters. In practice, γ is often fixed at either 1 or 2, and λ_n is selected as the primary tuning parameter. Thus, we discuss the case of a fixed $\gamma \geq 1$ and the initial estimator $\hat{\beta}^{\text{init}} = \hat{\beta}^{\text{ols}}$ because of the fixed p setting.

Alasso enjoys the oracle property in regressions with weakly dependent regressors (Medeiros and Mendes, 2016). The following Theorem 1 confirms that Alasso maintains the oracle property in regressions with unit root regressors. Let $\hat{M}^A = \{j : \hat{\beta}_j^A \neq 0\}$ be Alasso's estimated active set, and $\bar{\delta} = \max\{(\delta_j)_{j=1}^p\}$ be the fastest speed such that β_{jn}^* shrinks to 0.

Theorem 1. Suppose that the linear model (1) satisfies Assumption 1. If the tuning parameters λ_n and γ are chosen such that

$$\frac{\lambda_n}{n^{1-\bar{\delta}\gamma}} + \frac{1}{\lambda_n} \rightarrow 0 \quad (7)$$

and $\bar{\delta}\gamma < 1$, then

- (a) Variable selection consistency: $P(\hat{M}^A = M^*) \rightarrow 1$.
- (b) Asymptotic distribution: $n(\hat{\beta}^A - \beta_n^*)_{M^*} \Rightarrow \Omega_{M^*}^{-1} \zeta_{M^*}$.

As shown in Theorem 1(a), the estimated active set $\hat{M}^A$ coincides with the true active set M^* wpa1; in other words, $\hat{\beta}_j^A \neq 0$ if $j \in M^*$ and $\hat{\beta}_j^A = 0$ if $j \in M^{*c}$. (b) indicates that Alasso's asymptotic distribution in the true active set is as if the oracle M^* is known. In this nonstationary regression, Alasso's adaptiveness is maintained through the proper choice of $\hat{\tau}_j = |\hat{\beta}_j^{\text{ols}}|^{-\gamma}$. When the true coefficient is nonzero, $\hat{\tau}_j$ delivers a penalty of a negligible order $\lambda_n/n^{1-\bar{\delta}_j\gamma} \rightarrow 0$, recovering the OLS limit theory. When the true coefficient is zero, $\hat{\tau}_j$ imposes a heavier penalty of the order $\lambda_n/n^{1-\gamma} \rightarrow \infty$, thereby achieving consistent variable selection. The intuition in Zou (2006, Remark 2) under deterministic design is generalized in our proof of the setting with nonstationary regressors.

Both $\bar{\delta}$ and γ appear in the rate condition (7) for generality. In practice, because $\bar{\delta}$ is an unknown feature of the DGP and γ can be controlled by the user, we recommend setting $\gamma = 1$, which simplifies (7) as $\lambda_n/n^{1-\bar{\delta}} + 1/\lambda_n \rightarrow 0$. A conservative choice of λ_n diverging more slowly than polynomial orders of n , say, $\lambda_n \asymp \log \log n$, satisfies the above condition for all $\bar{\delta} > 0$. The practical choice of the tuning parameter in the numerical simulations and the empirical application will be discussed in the corresponding sections.

3. Adaptive LASSO with mixed roots

In practice, we often encounter a multitude of candidate predictors that exhibit various dynamic patterns. Some are stationary, whereas others can be highly persistent and/or cointegrated. In this section, we discuss the theoretical properties of Alasso under a mixed persistence environment. We extend the model in Section 2 to accommodate I(0) and (local) unit root regressors, with possible cointegration in the latter. The theory in this section provides general guidance for multivariate predictive regressions.

3.1. Model

We introduce three types of predictors into the model. A $1 \times p_c$ cointegrated system $x_i^c = (x_{i1}^c, \dots, x_{ip_c}^c)$ has cointegration rank p_1 , so that $p_2 = p_c - p_1$ is the number of local unit roots in this cointegration system. Let $x_i^c = (x_{i1}^c, x_{i2}^c)$ admit a triangular representation

$$\begin{aligned} x_{1i}^{c'} - A_1 x_{2i}^{c'} &= v_{1i}', \\ p_1 \times 1 \quad p_1 \times p_2 \quad p_2 \times 1 \quad p_1 \times 1 \end{aligned} \quad (8)$$

$$(I_{p_2} - R_2 L)x_{2i}^{c'} = v_{2i}',$$

where $R_2 = I_{p_2} + c_2/n$ with $c_2 = \text{diag}(\check{c}_1, \dots, \check{c}_{p_2})$, L is the lag operator, and the vector v_{1i} is the cointegrating residual, whose initialization is an $O_p(1)$ stationary variable. Each local-to-unity parameter $\check{c}_l$ is finite, thus including both stationary and nonstationary local unit root regions. This local-to-unity specification includes the unit root process as a special case when $\check{c}_l = 0$. The triangular representation (Phillips, 1991; Elliott, 1998) is a convenient and general form of cointegrated systems. Xu (2018) also adopts this structure in predictive regressions.

Assume that y_i is generated from the linear model

$$y_i = \sum_{l=1}^{p_z} z_{il}\alpha_l^* + \sum_{l=1}^{p_1} v_{1il}\phi_l^* + \sum_{l=1}^{p_x} x_{il}\beta_l^* + u_i = z_i\alpha^* + v_{1i}\phi_1^* + x_i\beta^* + u_i. \quad (9)$$

Each time series $z_l = (z_{1l}, \dots, z_{nl})'$ is a stationary regressor. Each $x_l = (x_{1l}, \dots, x_{nl})'$ is a local-to-unity process (initialized from $O_p(1)$ stationary process) such that $x_i = (x_{i1}, \dots, x_{ip_x})$ satisfies $(I_{p_x} - R_x L)x_i' = e_i'$, where $R_x = I_{p_x} + c_x/n$ with $c_x = \text{diag}(\check{c}_1, \dots, \check{c}_{p_x})$ for finite $\check{c}_l$.

Eq. (9) is *infeasible* because the cointegrating residual v_{1i} is unobservable without *a priori* knowledge about the cointegration relationship. What we observe is the vector x_i^c that contains cointegration groups. Substituting (8) into (9), we obtain a *feasible* regression equation

$$y_i = z_i\alpha^* + x_{1i}^c\phi_1^* + x_{2i}^c\phi_2^* + x_i\beta^* + u_i = z_i\alpha^* + x_i^c\phi^* + x_i\beta^* + u_i, \quad (10)$$

where $\phi_2^* = -A_1'\phi_1^*$ and $\phi^* = (\phi_1^*, \phi_2^*)'$. Stacking the sample of n observations, the infeasible and feasible regressions can be written as

$$y = Z\alpha^* + V_1\phi_1^* + X\beta^* + u \quad (11)$$

$$= Z\alpha^* + X_1^c\phi_1^* + X_2^c\phi_2^* + X\beta^* + u$$

$$= Z\alpha^* + X^c\phi^* + X\beta^* + u, \quad (12)$$

where the variable V_1 is the $n \times p_1$ matrix that stacks $(v_{1i})_{i=1}^n$, and X_1^c , X_2^c , X^c , Z , and X are defined similarly. We explicitly define $\alpha^* = \alpha^{0*}$ and $\phi^* = \phi^{0*}$ as two coefficients independent of the sample size, which are associated with Z and X^c , respectively. As in Section 2, the coefficients associated with X are specified as local-to-zero sequences $\beta^* = \beta_n^* = (\beta_l^{0*}/n^{\delta_l})_{l=1}^{p_x}$, where β_l^{0*} is invariant to the sample size.

We assume a linear process for the innovation and cointegrating residual vectors. In contrast to the simplistic iid assumption in Section 2, the linear process assumption is fairly general, including as special cases many practical dependent processes such as the stationary autoregressive and moving average processes. Let $v_i = (v_{1i}, v_{2i})$ and $p = p_z + p_c + p_x$.

Assumption 2 (Linear Process). The vector of stacked innovation and stationary predictors follows the linear process:

$$\xi_i := (z_i, v_i, e_i, u_i)' = F(L)\varepsilon_i = \sum_{k=0}^{\infty} F_k\varepsilon_{i-k},$$

where $\varepsilon_i = \begin{pmatrix} \varepsilon_i^{(z)'} \\ \varepsilon_i^{(v)'} \\ \varepsilon_i^{(e)'} \\ \varepsilon_i^{(u)'} \end{pmatrix} \sim \text{iid} \left(0, \Sigma_\varepsilon = \begin{pmatrix} \Sigma_{zz} & \Sigma_{zv} & \Sigma_{ze} & 0 \\ \Sigma'_{zv} & \Sigma_{vv} & \Sigma_{ve} & 0 \\ \Sigma'_{ze} & \Sigma'_{ve} & \Sigma_{ee} & \Sigma_{eu} \\ 0 & 0 & \Sigma'_{eu} & \Sigma_{uu} \end{pmatrix} \right), F_0 = I_{p+1}, \sum_{k=0}^{\infty} k \|F_k\| < \infty, F(x) = \sum_{k=0}^{\infty} F_k x^k$

and $F(1) = \sum_{k=0}^{\infty} F_k > 0$.

Remark 1. Following the cointegration and predictive regression literature, we allow the correlation between the innovation of the regression error $\varepsilon_i^{(u)}$ and the innovation of nonstationary predictors $\varepsilon_i^{(e)}$. However, to ensure identification, we rule out the correlation between $\varepsilon_i^{(u)}$ and the innovation of the stationary or cointegrated predictors.

3.2. OLS

As Allasso attaches a penalty term to the OLS criterion function, we first study the asymptotic distribution of the OLS estimator

$$\hat{\theta}^{\text{ols}} = (W'W)^{-1}W'y$$

under the mixed roots, where $W = (Z, X^c, X)$ is the observed predictor matrix. To state the result, we define the true coefficients $\theta_n^* = (\alpha^{0*}, \phi_1^{0*}, \phi_2^{0*}, \beta_n^{*'})'$, where $\phi_2^{0*} = -A_1'\phi_1^{0*}$, a diagonal normalizing matrix $R_n = \text{diag}((\sqrt{n}\mathbf{1}'_{p_z+p_1}, n\mathbf{1}'_{p_2+p_x}))$,

and a rotation matrix $Q = \begin{pmatrix} I_{p_z} & 0 & 0 & 0 \\ 0 & I_{p_1} & 0 & 0 \\ 0 & A_1' & I_{p_2} & 0 \\ 0 & 0 & 0 & I_{p_x} \end{pmatrix}.$

Theorem 2. If the linear model (10) satisfies Assumption 2, then

$$R_n Q (\hat{\theta}^{\text{ols}} - \theta_n^*) = \begin{pmatrix} \sqrt{n}(\hat{\alpha}^{\text{ols}} - \alpha^{0*}) \\ \sqrt{n}(\hat{\phi}_1^{\text{ols}} - \phi_1^{0*}) \\ n(A_1' \hat{\phi}_1^{\text{ols}} + \hat{\phi}_2^{\text{ols}}) \\ n(\hat{\beta}^{\text{ols}} - \beta_n^*) \end{pmatrix} \Rightarrow (\Omega^+)^{-1} \zeta^+. \quad (13)$$

where Ω^+ is the weak limit of $R_n^{-1} Q'^{-1} W' W Q^{-1} R_n^{-1}$ and ζ^+ is the weak limit of $R_n^{-1} Q'^{-1} W' u$, whose explicit expressions are spelled out in (A.14) and (A.15) in Appendix A.2, respectively.

Remark 2. To see the effect of rotation matrix Q , note that the definitions of Ω^+ and ζ^+ include the transformation $Q'^{-1} W = (Z^+, X^+)$, where $Z^+ = (Z, V_1)$ is the “extended” stationary regressors and $X^+ = (X_2^c, X)$ is the “extended” nonstationary regressors. The feasible regressors in W are mapped by Q'^{-1} into the infeasible regressors (Z^+, X^+) in which the stationary and nonstationary components are separated. Then, the scaling factor R_n^{-1} sends the building blocks of the regression analysis to their respective asymptotic weak limits as $n \rightarrow \infty$, in which Ω^+ is the weak limit of the Gram matrix and ζ^+ is the weak limit of the empirical process.

Remark 3. Because we keep an agnostic view about the identities of the stationary, local unit root, and cointegrated regressors, Theorem 2 is not useful for statistical inference, as we do not know which coefficients converge at the $\sqrt{n}$ -rate and which at the n -rate. (13) shows that with the help of the rotation Q , the estimators $\hat{\phi}_1^{\text{ols}}$ and $\hat{\phi}_2^{\text{ols}}$ are tightly connected in the sense that $A_1' \hat{\phi}_1^{\text{ols}} + \hat{\phi}_2^{\text{ols}} = O_p(n^{-1})$. Without the rotation that is unknown in practice, the OLS components associated with the stationary and cointegration systems converge at the $\sqrt{n}$ rate, whereas only those associated with the non-cointegration local unit roots converge at the n rate. Let $R_n^f = \text{diag}((\sqrt{n} \mathbf{1}_{p-p_x}, n \mathbf{1}_{p_x}))$ be the normalizing matrix for the feasible components; then,

$$R_n^f (\hat{\theta}^{\text{ols}} - \theta_n^*) \Rightarrow \left(\lim_{n \rightarrow \infty} R_n^f Q^{-1} R_n^{-1} \right) (\Omega^+)^{-1} \zeta^+,$$

or, more explicitly,

$$\begin{pmatrix} \sqrt{n}(\hat{\alpha}^{\text{ols}} - \alpha^{0*}) \\ \sqrt{n}(\hat{\phi}_1^{\text{ols}} - \phi_1^{0*}) \\ \sqrt{n}(\hat{\phi}_2^{\text{ols}} - \phi_2^{0*}) \\ n(\hat{\beta}^{\text{ols}} - \beta_n^*) \end{pmatrix} \Rightarrow \begin{pmatrix} I_{p_z} & 0 & 0 & 0 \\ 0 & I_{p_1} & 0 & 0 \\ 0 & -A_1' & 0 & 0 \\ 0 & 0 & 0 & I_{p_x} \end{pmatrix} (\Omega^+)^{-1} \zeta^+. \quad (14)$$

Although each variable in the cointegration system appears as a local-to-unity process, as a group they are extremely comoving, reducing their rate of convergence from n to $\sqrt{n}$. This effect is analogous to the deterioration of the convergence rate of the OLS estimator for nearly perfectly collinear regressors.

3.3. Adaptive LASSO

Similar to Section 2.2, Alasso for model (12) is estimated as

$$\hat{\theta}^A = \arg \min_{\theta} \left\{ \|y - W\theta\|^2 + \lambda_n \sum_{j=1}^p \hat{\tau}_j |\theta_j| \right\}, \quad (15)$$

where $\hat{\tau}_j = |\hat{\theta}_j^{\text{ols}}|^{-\gamma}$. The literature on Alasso has established the oracle property in many models. Caner and Knight (2013) and Kock (2016) study Alasso's rate adaptiveness in a pure autoregressive setting with iid error processes. In their cases, the potential nonstationary regressor is the first-order lagged dependent variable, and the other regressors are stationary. Therefore, the components of different convergence rates are known in advance. We complement this line of nonstationary LASSO literature by allowing a general regression framework with mixed degrees of persistence. We also generalize the error processes to the commonly used dependent processes, which is important in practice. For example, the long-horizon return regression in Section 6 requires this type of dependence in its error structure because of the overlapping return construction.

Surprisingly, Theorem 3 shows that in the mixed root model, Alasso's oracle property holds partially, but not for all regressors. To discuss variable selection in this context, we introduce the following notations. We partition the index set of all regressors $\mathcal{M} = \{1, \dots, p\}$ into four components: $\mathcal{I}_0$ ($I(0)$ variables associated with Z), $\mathcal{C}_1$ (associated with X_1^c), $\mathcal{C}_2$ (associated with X_2^c), and $\mathcal{I}_1$ (nonstationary variables associated with X). Let $\mathcal{C} = \mathcal{C}_1 \cup \mathcal{C}_2$ and $\mathcal{I} = \mathcal{I}_0 \cup \mathcal{I}_1$. Let $M^* = \{j : \theta_j^{0*} \neq 0\}$ be the true active set for the feasible representation, and let $\hat{M}^A = \{j : \hat{\theta}_j^A \neq 0\}$ be Alasso's estimated active set. Next, let $\mathcal{M}_Q = \mathcal{I} \cup \mathcal{C}_1$ be the set of coordinates that are invariant to the rotation in Q . Similarly, let

Table 1
Diagram of a cointegrating system in predictors.

$(\phi_1^{0*}, \phi_2^{0*})$ (c_1, c_2)	★ _a x_a^c	★ _b x_b^c	0 x_c^c	0 x_d^c	0 x_e^c	♣ _f x_f^c	♣ _g x_g^c	0 x_h^c	0 x_q^c	coint. resid.	ϕ_1^{0*}
	1	0	0	0	0	♣ _{fa}	♣ _{ga}	0	0	v_a	★ _a
	0	1	0	0	0	0	♣ _{gb}	0	0	v_b	★ _b
$(I_{p_1}, -A_1')$	0	0	1	0	0	0	0	♣ _{hc}	0	v_c	0
	0	0	0	1	0	♣ _{fd}	0	♣ _{hd}	♣ _{qd}	v_d	0
	0	0	0	0	1	0	♣ _{ge}	0	0	v_e	0

Note: The diagram represents a cointegration system of 9 variables $x_a^c, x_b^c, \dots$ of cointegrating rank 5. The last column represents the coefficients in ϕ_1^{0*} , with ★ as a nonzero entry. In the matrix $-A_1'$, ♣ is nonzero, and the coefficients in gray cells are irrelevant to the value $\phi_2^{0*} = -A_1' \phi_1^{0*}$ no matter zero or nonzero. The first row displays the coefficients ϕ_1^{0*} (the same as in the last column, with ★ for non-zeros) and ϕ_2^{0*} (♣ for non-zeros). In this example, $(x_a^c, x_b^c, x_f^c, x_g^c)$ are in the active set M^* . There are two active cointegration groups, (x_a^c, x_f^c, x_g^c) and (x_b^c, x_g^c) , and three inactive cointegration groups, (x_c^c, x_h^c) , $(x_d^c, x_f^c, x_g^c, x_q^c)$, and (x_e^c, x_g^c) .

$M_Q^* = M^* \cap \mathcal{M}_Q$ be the active set in the infeasible regression equation (9), and let $M_Q^{*c} = \mathcal{M} \setminus M_Q^*$ be the corresponding inactive set. The DGP (11) obviously implies $c_2 \cap M_Q^* = \emptyset$ and $c_2 \subseteq M_Q^{*c}$. For a generic index set $M \subseteq \mathcal{M}$, let $\text{CoRk}(M)$ be the cointegration rank of the variables in M .

Theorem 3. Suppose that the linear model (10) satisfies Assumption 2 and that for all of the coefficients in the set c_2 , we have

$$\phi_{2l}^{0*} \neq 0 \text{ if } \sum_{s=1}^{p_1} |A_{1sj} \phi_{1s}^{0*}| \neq 0. \quad (16)$$

If the tuning parameters λ_n and γ are chosen such that

$$\frac{\lambda_n}{n^{(1-\delta\gamma) \wedge 0.5}} + \frac{1}{\lambda_n} \rightarrow 0 \quad (17)$$

and $\bar{\delta}\gamma < 1$, then we have the following results:

(a) Consistency and asymptotic distribution:

$$(R_n Q(\hat{\theta}^A - \theta_n^*))_{M_Q^*} \implies (\Omega_{M_Q^*}^+)^{-1} \zeta_{M_Q^*}^+ \quad (18)$$

$$(R_n Q(\hat{\theta}^A - \theta_n^*))_{M_Q^{*c}} \xrightarrow{P} 0. \quad (19)$$

(b) Partial variable selection consistency:

$$P(M^* \cap \mathcal{I} = \hat{M}^A \cap \mathcal{I}) \rightarrow 1, \quad (20)$$

$$P((M^* \cap \mathcal{C}) \subseteq (\hat{M}^A \cap \mathcal{C})) \rightarrow 1, \quad (21)$$

$$P(\text{CoRk}(M^*) = \text{CoRk}(\hat{M}^A)) \rightarrow 1. \quad (22)$$

Remark 4. Condition (16) is an extra assumption that rules out the pathological case that some nonzero elements in $A_{1s} \phi_{1s}^{0*}$, $s = 1, \dots, p_1$, happen to exactly cancel out one another and render ϕ_{2l}^{0*} inactive. In other words, it ensures that if an x_l^c is involved in more than one active cointegration group, it must be active in (10). This condition holds in general, as it is violated only under very specific configurations of ϕ_1^{0*} and A_1 . For instance, in the demonstrative example in Table 1, the condition breaks down if x_g^c 's true coefficient $\clubsuit_{ga} \times \star_a + \clubsuit_{gb} \times \star_b = 0$.

Were we informed of the oracle about the true active variables in M^* and the cointegration matrix A_1 , we would transform the cointegrated variables into cointegrating residuals, discard the inactive variables, and then run OLS. Ideally, we would conduct an estimation with variables in M_Q^* only. Such an oracle OLS shares the same asymptotic distribution as its Allasso counterpart in (18). Outside of the active set M_Q^* , (19) shows that all of the other (transformed) variables in M_Q^{*c} consistently converge to zero.

The variable selection results in Theorem 3 are novel and interesting. (20) shows that variable selection is consistent for the pure $I(0)$ and local unit root variables, which is in line with the well-known oracle property of Allasso. However, instead of confirming the oracle property, (21) indicates that in the cointegration set $\mathcal{C}$, the selected $\hat{M}^A$ asymptotically contains the true active variables in M^* but Allasso may over-select inactive variables. The inconsistency stems from the mismatch

between the convergence rate of the initial estimator and the marginal behavior of a cointegrated variable viewed in isolation. Consider a pair of inactive cointegrating variables, such as (x_c^c, x_h^c) in Table 1. The unknown cointegration relationship precludes transforming this pair into the cointegrating residual v_c . Without the rotation, OLS associated with the pair can only achieve the $\sqrt{n}$ rate according to (14). The resulting penalty weights are insufficient to remove these variables that *individually* appear as nonstationary. In consequence, Alasso fails to eliminate *both* x_c^c and x_h^c wpa1. To the best of our knowledge, this is the first case of Alasso's variable selection inconsistency in an important empirical model.

Under condition (16) all variables in the active cointegration groups have nonzero coefficients, and Alasso selects them asymptotically according to (21). Despite potential variable over-selection in $\mathcal{C}$, (22) brings relief: in the limit, the cointegration rank in Alasso's selected set, $\text{CoRk}(\hat{M}^A)$, must equal the active cointegration rank $\text{CoRk}(M^*)$. Note that under our agnostic perspective, we do not need to know or use testing procedures to determine the value of $\text{CoRk}(M^*)$. Let $C^{*c} = M^{*c} \cap \mathcal{C}$ be the index set of the inactive variables in $\mathcal{C}$. (21) implies that the variables in $C^{*c} \cap \hat{M}^A$ —Alasso's mistakenly selected inactive variables—cannot form cointegration groups. In mathematical expression, $P(\text{CoRk}(C^{*c} \cap \hat{M}^A) = 0) \rightarrow 1$.

Example 1. Let us again take (x_c^c, x_h^c) in Table 1 as an example. (22) indicates that Alasso is at least partially effective in that it prevents the inactive x_c^c and x_h^c from entering $\hat{M}^A$ simultaneously, as it kills at least one variable in the pair to break the cointegration relationship. The intuition is as follows. Suppose that (x_c^c, x_h^c) are both selected. Because of coefficient estimation consistency, in the predictive regression these two variables together behave like the cointegrating residual v_c , which is an $I(0)$. In the limit Alasso will not tolerate this inactive v_c , because these coefficients associated with the underlying x_c and x_h are subject to a penalty weight $\hat{\tau}_j = 1/O_p(n^{-1/2})$, which is of the same order as the pure $I(0)$ variables. Recall that Alasso removes all inactive pure $I(0)$ variables when $n \rightarrow \infty$ under the same level of penalty weight.

Example 2. Similar reasoning applies in another inactive cointegration group (x_e^c, x_g^c) . In the regression, x_e^c is inactive but x_g^c is active, as it is involved in the active cointegration groups (x_a^c, x_f^c, x_g^c) and (x_c^c, x_h^c) . Clearly x_g^c is selected wpa1, but let us suppose that x_e^c is also selected. If so, x_e^c 's contribution to the predictive regression would be equivalent to the $I(0)$ cointegrating residual v_e . Its corresponding penalty is of the order $1/O_p(n^{-1/2})$, which is sufficient to remove the inactive v_e . Therefore, x_e^c cannot survive Alasso's variable selection.

The intuition gleaned in these examples can be generalized to cointegration relationships involving more than two variables and multiple cointegrated groups. The proof of Theorem 3 formalizes this argument by inspecting a linear combination of the corresponding Karush–Kuhn–Tucker (KKT) condition for the selected variables.

In the literature, Alasso achieved the oracle property in a single implementation, so there was no need to run it twice. In our predictive regression with mixed roots, it is possible that a single run of Alasso over-selects inactive variables in the cointegration system. Given that the cointegrating ties are all shattered wpa1 in (22), further action can help fulfill the oracle property.

3.4. Twin adaptive LASSO

When the sample size is sufficiently large, with high probability, the mistakenly selected inactive variables have no cointegration relationship, so they behave as isolated local-to-unity processes in the post-selection regression equation of y on $(w_j)_{j \in \hat{M}^A}$, where $w_j = (w_{1j}, \dots, w_{nj})'$ is the j th time series regressors. This observation suggests the need to run a post-selection Alasso. We first obtain the post-Alasso initial OLS estimator

$$\hat{\theta}^{\text{po}} = (W'_{\hat{M}^A} W_{\hat{M}^A})^{-1} W'_{\hat{M}^A} y.$$

The post-selection OLS estimator $\hat{\theta}_j^{\text{po}} = O_p(n^{-1})$ for the over-selected inactive cointegrated variables $j \in C^{*c} \cap \hat{M}^A$, rather than $O_p(n^{-1/2})$, as for the first-around initial $\hat{\theta}_j^{\text{ols}}$. The resulting post-selection penalty level $\hat{\tau}_j^{\text{po}} = |\hat{\theta}_j^{\text{po}}|^{-\gamma}$ is sufficiently heavy to wipe out these redundant variables in another round of Alasso. We call this procedure TALasso, for which $\hat{\theta}^{\text{TA}} = (\hat{\theta}_{\hat{M}^A}^{\text{TA}}, \hat{\theta}_{\mathcal{M} \setminus \hat{M}^A}^{\text{TA}})'$, where $\hat{\theta}_{\mathcal{M} \setminus \hat{M}^A}^{\text{TA}} = \mathbf{0}$ and

$$\hat{\theta}_{\hat{M}^A}^{\text{TA}} = \arg \min_{\theta} \left\{ \|y - \sum_{j \in \hat{M}^A} w_j \theta_j\|^2 + \lambda_n \sum_{j \in \hat{M}^A} \hat{\tau}_j^{\text{po}} |\theta_j| \right\}. \quad (23)$$

TAlasso asymptotically reclaims variable selection consistency for all types of variables. Let $\hat{M}^{\text{TA}}$ be the active set for TALasso.

Theorem 4. Under the same assumptions and the same rate for λ_n as in Theorem 3, the estimator $\hat{\theta}^{\text{TA}}$ satisfies

- (a) Asymptotic distribution: $(R_n Q(\hat{\theta}^{\text{TA}} - \theta_n^*))_{M_Q^*} \Rightarrow (\Omega_{M_Q^*}^+)^{-1} \zeta_{M_Q^*}^+$;
- (b) Variable selection consistency: $P(\hat{M}^{\text{TA}} = M^*) \rightarrow 1$.

Faced with a variety of potential predictors with unknown orders of integration, we may not be able to sort them into different persistence categories in predictive regressions without potential testing errors (Smeekes and Wijler, 2020). Our research provides a valuable practical algorithm to address this problem. TAlasso is the first estimator that achieves the desirable oracle property without requiring prior knowledge on the persistence of multivariate regressors. Despite the intensive study of LASSO in recent years, Theorems 3 and 4 are novel and important. With the cointegration system in the predictors, the former shows that Alasso does not automatically adapt to the behavior of a system of regressors. Nevertheless, it at least breaks all redundant cointegration groups so that its defects can be easily addressed by another round of Alasso. This solution echoes the repeated implementation of a machine learning procedure as in Phillips and Shi (2021).

4. Conventional LASSO with mixed roots

Originally, LASSO was proposed as a plain l_1 -penalized regression without a sophisticated weighting scheme, motivated by the optimization problem's variable screening effect: some coefficients are estimated as exactly zeros in finite samples under a wide range of the tuning parameter (Tibshirani, 1996). As mentioned in the introduction, in this paper, we call this original estimator

$$\hat{\theta}^P = \arg \min_{\theta} \{ \|y - W\theta\|^2 + \lambda_n \|\theta\|_1 \} \quad (24)$$

Plasso. However, Plasso is scale-variant in the sense that if we change the unit of w_j by multiplying it by a nonzero constant c , such a change is not reflected in the penalty term in (24), so the Plasso estimator does not change proportionally to $\hat{\theta}_j^P/c$. To keep the estimation scale-invariant to the choice of the unit of x_j , researchers often scale-standardize LASSO as

$$\hat{\theta}^S = \arg \min_{\theta} \left\{ \|y - W\theta\|^2 + \lambda_n \sum_{j=1}^p \hat{\sigma}_j |\theta_j| \right\}. \quad (25)$$

where $\hat{\sigma}_j = \sqrt{n^{-1} \sum_{i=1}^n (w_{ij} - \bar{w}_j)^2}$ is the sample standard deviation of w_j and $\bar{w}_j$ is the sample average. In this paper, we call (25) Slasso. Such standardization is the default option for LASSO in many statistical packages, such as the R package `glmnet`. Note that Alasso is also scale-invariant with $\hat{\theta}^{\text{init}} = \hat{\theta}^{\text{ols}}$ and $\gamma = 1$.

Both Plasso and Slasso are special cases of $\min_{\theta} \{ \|y - X\theta\|_2^2 + \lambda_n \sum_{j=1}^p \hat{\tau}_j |\theta_j| \}$, where the former uses $\hat{\tau}_j = 1$ and the latter specifies $\hat{\tau}_j = \hat{\sigma}_j$. When Slasso's scale standardization is carried out with stationary and weakly dependent regressors, each $\hat{\sigma}_j^2$ converges in probability to its finite population variance. In the literature, the difference between the asymptotic properties of Plasso and Slasso are not of interest because Slasso uses another set of constant configurations for the penalty levels.

In sharp contrast, for those w_j associated with $\mathcal{C} \cup \mathcal{I}_1$ that are individually nonstationary, by (4), we have

$$\frac{\hat{\sigma}_j}{\sqrt{n}} = \sqrt{\frac{1}{n^2} \sum_{i=1}^n (w_{ij} - \bar{w}_j)^2} \implies d_j, \quad (26)$$

where d_j is a non-degenerate random variable (whose expression can be found in the proof of Corollary 2), so that the sample standard deviation $\hat{\sigma}_j = O_p(\sqrt{n})$ diverges in the limit. As a result, it imposes a much heavier penalty on the associated coefficients than the stationary time series does.

Adopting a standard argument for LASSO as in Knight and Fu (2000), we derive the following results characterizing Plasso's asymptotic behavior with various choices of λ_n . For exposition, we define a function $D : \mathbb{R}^3 \mapsto \mathbb{R}$ as $D(s, v, b) = s[v \cdot \text{sgn}(b)I(b \neq 0) + |v|I(b = 0)]$, where $\text{sgn}(b) = I(b > 0) - I(b < 0)$ is the sign function.

Corollary 1. Suppose that the linear model (10) satisfies Assumption 2.

- (a) If $\lambda_n \rightarrow \infty$ and $\lambda_n/\sqrt{n} \rightarrow 0$, then $R_n Q(\hat{\theta}^P - \theta_n^*) \implies (\Omega^+)^{-1} \zeta^+$.
- (b) If $\lambda_n/\sqrt{n} \rightarrow c_\lambda \in (0, \infty)$, then

$$R_n Q(\hat{\theta}^P - \theta_n^*) \implies \arg \min_v \left\{ v' \Omega^+ v - 2v' \zeta^+ + c_\lambda \sum_{j \in \mathcal{I}_0 \cup \mathcal{C}} D(1, v_j, \theta_j^{0*}) \right\}.$$

- (c) If $\lambda_n/\sqrt{n} \rightarrow \infty$ and $\lambda_n/n \rightarrow 0$, then

$$\frac{1}{\lambda_n} R_n Q(\hat{\theta}^P - \theta_n^*) \implies \arg \min_v \left\{ v' \Omega^+ v + \sum_{j \in \mathcal{I}_0 \cup \mathcal{C}} D(1, v_j, \theta_j^{0*}) \right\}.$$

In Corollary 1(a), the tuning parameter is small and the limit distribution of Plasso is equivalent to that of OLS, so there is no variable screening effect. Similar to Plasso in the cross-sectional setting, there is no guarantee of consistent

variable selection. The screening effect occurs when the tuning parameter λ_n becomes larger. In view of the OLS rate of convergence in (14), we can call those θ_j associated with $\mathcal{I}_0 \cup \mathcal{C}$ the *slow coefficients* (at rate $\sqrt{n}$) and those associated with $\mathcal{I}_1$ the *fast coefficients* (at rate n). When λ_n is raised to the magnitude in (b), the term $D(1, v_j, \theta_j^{0*})$ causes variable screening among the slow coefficients but not the fast coefficients. If we further increase λ_n to the level of (c), then the convergence rate of the slow coefficients is dragged down by the large penalty, but there is still no variable screening effect for the fast coefficients. The local-to-zero coefficient δ_l for those in $\mathcal{I}_1$ plays no role in these asymptotic results because $\hat{\beta}_l$ converges to β_{ln}^* at the fast rate n regardless of δ_l .

Corollary 1 reveals a major drawback of Plasso in the mixed root model. Because it has one uniform penalty level for all variables, it is not adaptive to these various types of predictors. To induce variable screening in $\hat{\theta}_{\mathcal{I}_1}^p$, the tuning parameter must be ballooned to $\lambda/n \rightarrow c_\lambda \in (0, \infty]$, but the consistency of the slow coefficients would collapse under such a disproportionately heavy λ_n .

Remark 5. When nonstationary regressors are present, the results in Zou (2006, Section 2) are no longer applicable. Consider the simple case $W = X$ such that all of the regressors are non-cointegrating local unit roots. Recall that the random matrix Ω , defined in the line following (5), is the weak limit of $n^{-2}X'X$. Corollary 1(c) implies that in this case, the limiting distribution of $\arg \min_v \{v' \Omega v + \sum_{j=1}^p D(1, v_j, \theta_j^{0*})\}$ is non-degenerate because of the randomness of Ω . This differs from Lemma 3 of Zou (2006), in which the Plasso estimator degenerates to a constant. The distinction arises because in our context Ω is random, whereas in Zou's (2006) iid setting the counterpart of Ω degenerates to a non-random matrix.

We now turn to Slasso. Unlike the behavior of the slow coefficients in Plasso, the penalty scheme of Slasso makes the cointegration components in $\hat{\theta}^S$ susceptible to variable screening at much smaller scales of the tuning parameter.

Corollary 2. Suppose that the linear model (10) satisfies Assumption 2.

- (a) If $\lambda_n \rightarrow 0$, then $R_n Q(\hat{\theta}^S - \theta_n^*) \Rightarrow (\Omega^+)^{-1} \zeta^+$.
 (b) If $\lambda_n \rightarrow c_\lambda \in (0, \infty)$, then

$$R_n Q(\hat{\theta}^S - \theta_n^*) \Rightarrow \arg \min_v \left\{ v' \Omega^+ v - 2v' \zeta^+ + c_\lambda \sum_{j \in \mathcal{C}} D(d_j, v_j, \theta_j^{0*}) \right\}.$$

- (c) When $\lambda_n \rightarrow \infty$ and $\lambda_n/\sqrt{n} \rightarrow 0$, then

$$\frac{1}{\lambda_n} R_n Q(\hat{\theta}^S - \theta_n^*) \Rightarrow \arg \min_v \left\{ v' \Omega^+ v + \sum_{j \in \mathcal{C}} D(d_j, v_j, \theta_j^{0*}) \right\}.$$

Remark 6. The tuning parameter λ_n in Corollary 2 is an $\sqrt{n}$ order smaller than that in Corollary 1. Part(a) produces the same asymptotic distribution as OLS. The distinction of Plasso and Slasso arises from the coefficients in the set $\mathcal{C}$. Their corresponding penalty terms have the multipliers $\hat{\sigma}_j = O_p(\sqrt{n})$ rather than the desirable $O_p(1)$ that is suitable for their slow convergence rate under OLS. In other words, the penalty level is overly heavy for these parameters. The overwhelming penalty level causes a variable screening effect in (b) as soon as $\lambda_n \rightarrow c_\lambda \in (0, \infty)$. While the first argument of $D(\cdot, v_j, \theta_j^{0*})$ in Plasso is 1, it is replaced in Slasso by the random variable d_j , which introduces an extra source of uncertainty in variable screening. Moreover, (c) implies that for the consistency of $\hat{\phi}^S$, the tuning parameter λ_n must be small enough in the sense $\lambda_n/\sqrt{n} \rightarrow 0$; otherwise, they will be inconsistent. In both (b) and (c) the penalty term $D(d_j, v_j, \theta_j^{0*})$ screens those variables in $\mathcal{C}$ only. Again, the local-to-zero coefficient δ_l for those in $\mathcal{I}_1$ are irrelevant asymptotically.

To summarize this section, neither Plasso nor Slasso induces consistent parameter estimation and variable screening effect simultaneously for all components of the mixed regressors. The weight $\hat{\tau}_j$ is a constant for Plasso, while it exploits only the marginal variation of w_j for Slasso. Plasso uses a single tuning parameter and does not adapt to the different orders of magnitude of the slow and fast coefficients. Slasso suffers from overwhelming penalties for coefficients associated with the cointegration groups. Because the cointegrated regressors are individually local unit root processes, they can only form a linear combination to produce a stationary time series when classified into a system. In contrast, the penalty of Alasso/Talasso reflects the cointegration system via the initial/post-selection OLS.

5. Simulations

In this section, we examine via simulations the performance of the LASSO methods in forecasting and variable screening. We consider different sample sizes to demonstrate the quality of the asymptotic approximation in finite samples. Comparison is based on the one-period-ahead out-of-sample forecast.

5.1. Simulation design

Following the settings in Sections 2, 3, and 4, we consider four DGPs. Each DGP starts after a burn-in of 1000 periods for the innovations to mimic stationary $O_p(1)$ realizations.

DGP 1 (Pure unit roots). This DGP corresponds to the pure unit root model in Section 2. Consider a linear model with 9 unit root predictors, the same number as the persistent regressions in Section 6. Each regressor x_{ij} is drawn from the random walk $x_{ij} = x_{i-1,j} + e_{ij}$, and the dependent variable y_i is generated from $y_i = \gamma^* + x_i \beta_n^* + u_i$, where the innovations $(e_i, u_i) \sim \text{iid } N(0, \Sigma)$. The intercept $\gamma^* = 0.25$, the slope coefficient $\beta_n^* = (1, 1, 1, 0, 0, 0, 0, 0, 0)' / \sqrt{n}$, and the covariance matrix Σ is estimated from the data of Welch and Goyal (2008), as detailed in Supplement S1.

DGP 2 (Mixed roots and cointegration). This DGP is designed for the mixed root model in Section 3, which emulates the kitchen-sink approach in Section 6. The dependent variable

$$y_i = \gamma^* + \sum_{l=1}^3 \alpha_l^* z_{il} + \sum_{l=1}^4 \phi_l^* x_{il}^c + \sum_{l=1}^5 \beta_{ln}^* x_{il} + u_i, \quad (27)$$

where $\gamma^* = 0.3$, $\alpha^* = (0.4, 0, 0)$, $\phi^* = (0.5, -0.5, 0, 0)$, and $\beta_n^* = (1/\sqrt{n}, 1/\sqrt{n}, 0, 0, 0)$. The vector $\xi_i = (z_i, v_i, e_i, u_i)$ follows a VAR(1) process $\xi_i = \Phi \xi_{i-1} + \varepsilon_i$, where $\varepsilon_i \sim \text{iid } N(0, \Sigma_\varepsilon)$ in which Φ and Σ_ε are empirically estimated using the data of Welch and Goyal (2008), as detailed in Supplement S1. $x_i^c \in \mathbb{R}^4$ is a vector I(1) process with cointegration rank 2 based on the VECM, $\Delta x_i^c = \Gamma' \Lambda x_{i-1}^c + v_i$, where $\Lambda = \begin{pmatrix} 1 & -1 & 0 & 0 \\ 0 & 0 & 1 & -1 \end{pmatrix}$ and $\Gamma = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$ are the cointegrating matrix and the loading matrix, respectively. $(x_{il})_{l=1}^5$ are non-cointegrating random walks generated by $x_{il} = x_{i-1,l} + e_{il}$.

DGP 3 (Mixed roots with local-to-unity predictors) The coefficients and the innovation processes are the same as those in DGP 2, but we replace the I(1) processes in DGP 2 with local-to-unity processes. x_i^c is a vector of a local-to-unity process with cointegration rank 2 based on the triangular representation

$$\begin{pmatrix} 1 & -1 & 0 & 0 \\ 0 & 0 & 1 & -1 \end{pmatrix} x_i^c = \begin{pmatrix} v_{i2} \\ v_{i4} \end{pmatrix}$$

$$\left(I_2 - \text{diag} \left(\left(1 - \frac{4.8386}{n}, 1 \right) \right) \right) L \begin{pmatrix} x_{i1}^c \\ x_{i3}^c \end{pmatrix} = \begin{pmatrix} v_{i1} \\ v_{i3} \end{pmatrix}.$$

$(x_{il})_{l=1}^5$ are non-cointegrating local-to-unity processes generated by $x_{il} = (1 - \check{c}_l/n) x_{i-1,l} + e_{il}$, $l = 1, 2, \dots, 5$, and $(\check{c}_1, \check{c}_2, \dots, \check{c}_5) = (7.4934, 5.9152, 8.9277, 0, 0)$, where these local-unity coefficients are also estimated following Xu (2018). With some local-to-unity parameters being zero, DGP 3 incorporates a mixture of I(0), I(1), and local-to-unity processes.

DGP 4 (Stationary autoregression). The autoregressive distributed lag (ARDL) model is a classical specification for time series regressions. In addition to including lags of y_i , it is common to accommodate lags of predictors in predictive regressions, such as in Medeiros and Mendes (2016). The stationary dependent variable in the following equation is generated from the ARDL model

$$y_i = \gamma^* + \rho^* y_{i-1} + \sum_{l=1}^4 (\phi_{l1}^* x_{il}^c + \phi_{l2}^* x_{i-1,l}^c) + \sum_{l=1}^5 (\beta_{l1n}^* x_{il} + \beta_{l2n}^* x_{i-1,l}) + \sum_{l=1}^3 (\alpha_{l1}^* z_{il} + \alpha_{l2}^* z_{i-1,l}) + u_i,$$

where $\gamma^* = 0.3$, $\rho^* = 0.4$, $\phi_1^* = (0.75, -0.75, 0, 0)$, $\phi_2^* = (0, 0, 0, 0)$, $\beta_{1n}^* = (1.5/\sqrt{n}, 1, 0, 1/\sqrt{n}, 0)$, $\beta_{2n}^* = (0, -1, 0, 0, 0)$, $\alpha_1^* = (0.6, 0.4)$, $\alpha_2^* = (0.8, 0)$, and $\alpha_3^* = (0, 0)$. The predictors x_i^c , x_i , and z_i are generated in the same way as in DGP 3.

Remark 7. Our theory allows the local-to-zero coefficients for $\mathcal{I}_1$ to be $\beta_{ln}^* = \beta_l^{0*}/n^{\delta_l}$ for any $\delta_l \in (0, 1)$. In the simulations, we fix $\delta_l = 0.5$ for all $l \in \mathcal{I}_1$, which is the knife-edge rate for the balance of the predictive regression. When $\beta_l^{0*} \neq 0$, an active pure local unit root variable contributes to the dependent variable with

$$\beta_{ln}^* x_{\lfloor nr \rfloor, l} = \beta_l^{0*} (x_{\lfloor nr \rfloor, l} / \sqrt{n}) \implies \beta_l^{0*} J_{c_{\delta_l}}(r) = O_p(1),$$

for $r \in [0, 1]$, where $J_{c_{\delta_l}}(r)$ is an Ornstein–Uhlenbeck process. The limit process is of the same stochastic order of the active stationary variables in Z^+ , and so is y . In contrast, when $\delta_l \in (0.5, 1)$, we have $\beta_{ln}^* x_{\lfloor nr \rfloor, l} = (\beta_l^{0*}/n^{\delta_l-0.5}) (x_{\lfloor nr \rfloor, l} / \sqrt{n}) = o_p(1)$, and it is asymptotically dominated by the active stationary variables and the innovation u_i . Allasso/TAllasso would require a very large sample size to detect such weak signal in reality. When $\delta_l \in (0, 0.5)$, the active stationary variables in Z^+ are dominated by $\beta_{ln}^* x_{\lfloor nr \rfloor, l} = O_p(n^{0.5-\delta_l})$. In this case, the stochastic order of y is determined by the active variable in $\mathcal{I}_1$ of the largest order, and therefore $y_n = O_p(n^{0.5-\delta}) \rightarrow \infty$, where $\delta = \min\{\{\delta_l\}_{l \in \mathcal{I}_1}\}$. This violates the consensus that stock returns in most predictive regressions are stochastically bounded. Thus, we only consider $\delta_l = 0.5$ in the simulations.

As we develop our theory with regressors of fixed dimension, OLS is a natural benchmark. Another benchmark is the oracle OLS estimator under infeasible information. The sample sizes in our exercise are $n = 80, 120, 200, 400$, and 800 . We run 10,000 replications for each sample size and each DGP.

Each shrinkage estimator relies on its tuning parameter λ_n , which is the appropriate rate multiplied by a constant c_λ . We use fivefold cross validation (CV), where the sample is temporally ordered and then partitioned into five consecutive blocks, to guide the choice of c_λ . To ensure that the tuning parameter changes according to the rate specified in the asymptotic theory, we set $n = 100$ and run an exploratory simulation 100 times for each method that requires a tuning parameter. In each replication, we use the fivefold CV to obtain $c_\lambda^{(1)}, \dots, c_\lambda^{(100)}$. We then fix $c_\lambda = \text{median}(c_\lambda^{(1)}, \dots, c_\lambda^{(100)})$ in the full-scale 10,000 replications. To specify the tuning parameters for other sample sizes, we multiply the constant c_λ that we have calibrated from $n = 100$ by the rates suggested by our asymptotic theory. We multiply c_λ by $\sqrt{n}$ for Plasso and Slasso, and by $\sqrt{n}/(\log \log n)$ for Alasso and TAlasso.

5.2. Performance comparison

Table 2(a) reports the out-of-sample prediction accuracy in terms of the mean prediction squared error (MPSE) $E[(y_n - \hat{y}_n)^2]$. By the simulation design, unpredictable variation arises from the variance of the idiosyncratic error u_i , which is 1.

Plasso and Slasso achieve variable screening and consistent estimation, as the predictors are unit root processes in DGP 1. They are slightly better in forecasting than Alasso/TAlasso when the sample size is small; however, as the sample size increases to $n = 800$, TAlasso surpasses the conventional LASSO estimators, which suggests that variable selection is conducive to forecasting in a large sample. In DGPs 2–4 of mixed roots and cointegrated regressors, the settings are more difficult for the conventional LASSO to navigate. TAlasso is the best performer, followed by Alasso, which also outperforms the conventional LASSO methods by a non-trivial margin.

Table 2(b) summarizes the variable screening performance. Recall that $M^* = \{j : \theta_j^* \neq 0\}$ is the set of relevant regressors and $\hat{M} = \{j : \hat{\theta}_j \neq 0\}$ is the estimated active set. We define two *success rates* for variable screening:

$$SR_1 = |M^*|^{-1} E[|M^* \cap \hat{M}|], \quad SR_2 = |M^{*c}|^{-1} E[|M^{*c} \cap \hat{M}^c|]. \quad (28)$$

Here, SR_1 is the percentage of correct selections from the active set, and SR_2 is the percentage of correct removals from the inactive set. We also report the overall success rate of classification into zero or nonzero coefficients, defined as

$$SR = p^{-1} E\left[\left|\left\{j : I(\theta_j^* = 0) = I(\hat{\theta}_j = 0)\right\}\right|\right]. \quad (29)$$

These expectations in SR , SR_1 , and SR_2 are computed by the average across the simulation replications.

In terms of the overall selection measure SR , TAlasso is the most effective and Alasso takes the second place. As the sample size increases, TAlasso's success rates approach 100% in all DGPs, which supports variable selection consistency. While the difference in SR_1 among these methods becomes negligible when the sample size is large, the gain of TAlasso and Alasso stems largely from SR_2 . The asymptotic theory suggests that $\lambda_n \asymp \sqrt{n}$ is too small for Plasso to eliminate 0 coefficients corresponding to the nonstationary regressors. Plasso and Slasso achieve high SR_1 at the cost of low SR_2 . In view of the results in Table 2, the advantage of TAlasso's variable screening capability helps with forecast accuracy in the context of predictive regressions where many included regressors actually exhibit no predictive power.¹

Finally, as shown in Table 3, we check the variable screening of the inactive cointegration group (x_{i3}^c, x_{i4}^c) in DGPs 2–4, where the true coefficients $\phi_3^{0*} = \phi_4^{0*} = 0$. Alasso is effective in preventing both redundant variables from remaining in the regression according to the third column, while in the second column the chance of omitting one variable is around 40%–50%, indicating that it breaks the cointegration pair but cannot screen out both inactive regressors simultaneously. In contrast, TAlasso successfully identifies both redundant variables with significantly higher probabilities, as in the first column. Plasso and Slasso break down in variable selection consistency, as our theory suggests.

6. Empirical application

6.1. Data

We apply the LASSO methods to Welch and Goyal (2008)'s dataset to predict stock returns. We focus on the improvement in terms of prediction error and variable screening.

The dataset of Welch and Goyal (2008) is one of the most widely used in predictive regressions. Koo et al. (2020) update this monthly data from January 1945 to December 2012, and we use the same time span. The dependent variable is *excess return* (ExReturn), defined as the difference between the continuously compounded return on the S&P 500 index and the three-month Treasury bill rate. The estimated AR(1) coefficient of the excess return is 0.149, indicating weak

¹ In Table 2(b) Slasso has lower SR_2 than Plasso in a few instances. However, due to the presence of $\hat{\tau}_j = \hat{\sigma}_j = O_p(\sqrt{n})$ in the penalty term, in asymptotics it imposes a heavier penalty on coefficients of $I(1)$ regressors than Plasso does. The reason is that in our simulations we fix c_λ^p (for Plasso) and c_λ^s (for Slasso) by CV separately. CV selects the tuning parameter c_λ favoring lower MPSE and adjusts c_λ in finite sample. For example, in DGP 2, $c_\lambda^p = 0.6576$ is much larger than $c_\lambda^s = 0.1676$. If we fixed c_λ^p by CV and set $c_\lambda^s = c_\lambda^p$, Slasso would have a much higher SR_2 .

Table 2
MPSE and variable screening in simulations.

(a) MPSE (Relative to OLS)													
	<i>n</i>	Oracle	OLS	Alas.	TAlas.	Plas.	Slas.						
DGP 1	80	0.8614	1.0000	0.9256	0.9450	0.8945	0.8953						
	120	0.9090	1.0000	0.9622	0.9657	0.9347	0.9390						
	200	0.9425	1.0000	0.9900	0.9896	0.9604	0.9652						
	400	0.9699	1.0000	0.9976	0.9905	0.9861	0.9862						
	800	0.9841	1.0000	0.9977	0.9912	0.9935	0.9931						
DGP 2	80	0.8537	1.0000	0.9351	0.9295	0.9467	0.9651						
	120	0.8955	1.0000	0.9514	0.9447	0.9663	0.9831						
	200	0.9344	1.0000	0.9649	0.9615	0.9775	0.9985						
	400	0.9717	1.0000	0.9900	0.9854	0.9952	1.0145						
	800	0.9875	1.0000	0.9932	0.9906	0.9986	1.0238						
DGP 3	80	0.8452	1.0000	0.9247	0.9183	0.9457	0.9519						
	120	0.9046	1.0000	0.9562	0.9521	0.9746	0.9820						
	200	0.9373	1.0000	0.9705	0.9678	0.9826	0.9951						
	400	0.9701	1.0000	0.9887	0.9831	0.9954	1.0097						
	800	0.9874	1.0000	0.9927	0.9904	0.9990	1.0205						
DGP 4	80	0.7188	1.0000	0.8896	0.8766	0.9237	0.9229						
	120	0.8223	1.0000	0.9244	0.9129	0.9590	0.9765						
	200	0.8914	1.0000	0.9497	0.9423	0.9818	1.0225						
	400	0.9513	1.0000	0.9727	0.9686	0.9921	1.0431						
	800	0.9796	1.0000	0.9897	0.9866	0.9997	1.0593						
(b) Variable screening success rates													
	<i>n</i>	SR				SR ₁				SR ₂			
		Alas.	TAlas.	Plas.	Slas.	Alas.	TAlas.	Plas.	Slas.	Alas.	TAlas.	Plas.	Slas.
DGP 1	80	0.729	0.733	0.697	0.714	0.613	0.577	0.790	0.808	0.787	0.812	0.651	0.667
	120	0.764	0.769	0.688	0.731	0.674	0.644	0.862	0.863	0.809	0.831	0.602	0.664
	200	0.815	0.822	0.667	0.747	0.762	0.741	0.928	0.926	0.841	0.862	0.536	0.658
	400	0.879	0.887	0.623	0.763	0.865	0.856	0.976	0.974	0.886	0.903	0.446	0.658
	800	0.936	0.945	0.566	0.770	0.944	0.942	0.995	0.994	0.932	0.947	0.352	0.658
DGP 2	80	0.779	0.804	0.643	0.572	0.853	0.840	0.934	0.943	0.726	0.779	0.435	0.308
	120	0.820	0.846	0.634	0.584	0.897	0.889	0.967	0.968	0.765	0.816	0.397	0.309
	200	0.861	0.890	0.617	0.593	0.939	0.934	0.988	0.986	0.805	0.858	0.351	0.313
	400	0.905	0.936	0.593	0.601	0.977	0.976	0.998	0.997	0.853	0.908	0.303	0.318
	800	0.937	0.970	0.576	0.606	0.996	0.995	1.000	1.000	0.896	0.952	0.273	0.324
DGP 3	80	0.773	0.796	0.642	0.601	0.827	0.815	0.903	0.923	0.735	0.783	0.455	0.371
	120	0.812	0.836	0.637	0.614	0.872	0.864	0.947	0.952	0.768	0.816	0.416	0.373
	200	0.854	0.880	0.624	0.622	0.918	0.913	0.978	0.974	0.808	0.856	0.372	0.370
	400	0.901	0.931	0.605	0.633	0.967	0.965	0.996	0.994	0.854	0.907	0.326	0.375
	800	0.935	0.967	0.588	0.632	0.992	0.992	1.000	0.999	0.894	0.949	0.294	0.369
DGP 4	80	0.746	0.803	0.587	0.599	0.867	0.845	0.925	0.907	0.666	0.775	0.362	0.395
	120	0.782	0.845	0.587	0.616	0.903	0.890	0.950	0.924	0.701	0.815	0.344	0.411
	200	0.811	0.878	0.582	0.633	0.921	0.913	0.960	0.932	0.738	0.855	0.330	0.433
	400	0.836	0.908	0.576	0.648	0.933	0.930	0.966	0.927	0.772	0.892	0.316	0.463
	800	0.854	0.927	0.565	0.655	0.937	0.936	0.969	0.912	0.799	0.922	0.296	0.484

Note: Panel (a) compares the out-of-sample prediction accuracy in terms of MPSE $E[(y_n - \hat{y}_n)^2]$, where that of OLS is normalized to be 1. Panel (b) compares the variable screening performance in terms of the success rates SR, SR₁, and SR₂ defined in (28) and (29). Bold numbers indicate the best performance in each category of measurement.

persistence. The 12 financial and macroeconomic predictors are introduced and depicted in Fig. 1. Three variables, namely *ltr*, *inf1*, and *svar*, oscillate around the mean, and nine variables are highly persistent with estimated AR(1) coefficients greater than 0.95. The pairs (*tms*, *dfr*) and (*dp*, *dy*) are visibly moving in a synchronized pattern that suggests potential cointegration, and it is difficult to determine whether cointegration holds among (*dp*, *dy*, *ep*). *ep* fluctuates with *dy* before 2000, but the link dissolves after 2000, and the two series diverge in opposite directions during the Great Recession. The presence of stationary and persistent predictors fits the mixed roots environment studied in this paper, and our agnostic approach avoids decision errors resulting from hypothesis testing.

As recognized in the literature, the signal of persistent predictors may become stronger in long-horizon return predictions (Cochrane, 2009). In addition to the one-month-ahead short-horizon forecast, we construct the long-horizon excess return as the sum of the continuous compounded monthly excess return on the S&P 500 index

$$\text{LongReturn}_i = \sum_{k=i}^{i+12h-1} \text{ExReturn}_k,$$

Table 3Variable screening in DGPs 2–4 for the inactive cointegrated group $\phi_3^{0*} = \phi_4^{0*} = 0$.

	n	Both $\hat{\phi}_3, \hat{\phi}_4 = 0$				One and only one of $\hat{\phi}_3, \hat{\phi}_4 = 0$				Neither $\hat{\phi}_3, \hat{\phi}_4 = 0$			
		Alas.	TAlas.	Plas.	Slas.	Alas.	TAlas.	Plas.	Slas.	Alas.	TAlas.	Plas.	Slas.
DGP 2	80	0.443	0.597	0.182	0.181	0.480	0.344	0.572	0.541	0.077	0.059	0.246	0.277
	120	0.485	0.665	0.150	0.194	0.459	0.292	0.578	0.566	0.055	0.043	0.272	0.240
	200	0.529	0.738	0.112	0.214	0.436	0.234	0.566	0.581	0.036	0.028	0.322	0.205
	400	0.557	0.827	0.070	0.232	0.425	0.160	0.561	0.636	0.018	0.014	0.369	0.132
	800	0.603	0.907	0.050	0.273	0.392	0.090	0.551	0.645	0.006	0.004	0.399	0.082
DGP 3	80	0.424	0.596	0.144	0.241	0.491	0.338	0.624	0.604	0.085	0.066	0.232	0.156
	120	0.461	0.663	0.120	0.258	0.483	0.294	0.622	0.619	0.057	0.043	0.258	0.123
	200	0.499	0.734	0.089	0.262	0.467	0.239	0.602	0.637	0.034	0.028	0.309	0.102
	400	0.532	0.823	0.059	0.287	0.451	0.163	0.606	0.649	0.017	0.014	0.335	0.064
	800	0.568	0.907	0.042	0.314	0.427	0.089	0.591	0.634	0.005	0.004	0.368	0.052
DGP 4	80	0.326	0.561	0.156	0.343	0.526	0.366	0.510	0.510	0.149	0.073	0.334	0.148
	120	0.381	0.658	0.151	0.390	0.514	0.305	0.503	0.495	0.105	0.037	0.347	0.115
	200	0.418	0.740	0.134	0.409	0.506	0.240	0.496	0.508	0.076	0.020	0.371	0.083
	400	0.465	0.838	0.117	0.475	0.490	0.155	0.483	0.473	0.045	0.007	0.400	0.051
	800	0.499	0.903	0.099	0.509	0.476	0.096	0.457	0.459	0.025	0.002	0.444	0.032

Note: The numbers report the proportion of occurrences, over the replications, of eliminating both inactive variables (the first column, desirable outcome), eliminating only one variable (the second column), and eliminating neither variable (the third column).

where h is the length of the forecasting horizon, and $h = 1$ indicates one year. We use $h = 1/12$ (one month), $1/4$ (three months), $1/2$ (half a year), 1, 2, and 3 in our empirical exercises.

6.2. Performance

We forecast short- and long-horizon stock returns recursively with either a 10-year or 15-year rolling window. All 12 variables are made available in the predictive regression, which [Welch and Goyal \(2008\)](#) call the *kitchen sink model*. The tuning parameters for the shrinkage estimators are determined by a 10-fold CV on MPSE with consecutive partitions in each estimation window. The CV method is a preferred choice for prediction purposes. As a robustness check, we also use the Bayesian information criterion (BIC) to decide the tuning parameters for Alasso and TAlasso.²

The forecast returns of the LASSO methods are shown in [Fig. 2](#) along with the true realized return in gray for $h = 1/12 = 0.083$ (short horizon), 1 (median horizon), and 3 (long horizon). When $h = 0.083$, the realized excess return resembles white noise that is extremely difficult to forecast. When the horizon is extended to $h = 3$, the dynamics of the long-run aggregated return become clearer. For most of the time, Alasso and TAlasso track the realized return closely.

[Table 4](#) quantifies the forecast error in terms of the out-of-sample RMPSE (root MPSE) and mean predicted absolute error (MPAE) $E[y_n - \hat{y}_n]$. In addition to OLS, which includes all of the variables without any screening, we include random walk with drift (RWwD), i.e., the historical average of the excess returns, $\hat{y}_n = (n-1)^{-1} \sum_{i=1}^{n-1} y_i$, as another benchmark that utilizes no information from regressors. The results show that OLS loses in the short horizon, whereas RWwD suffers in the long horizon, indicating the ineffectiveness of the all-in and all-out approaches. Variable screening is essential for balanced performance in this empirical example. Alasso/TAlasso forecasts are more precise than Plasso/Slaslo in general. When the horizon is $h = 2$ or $h = 3$, TAlasso can achieve the smallest RMPSE, and Alasso is stronger than Plasso and Slaslo by a substantial margin. The results are robust when the tuning parameters are chosen by either CV or BIC.

There is an exceptional case of $h = 1$ with a 15-year rolling window. In this particular instance, Alasso fails to foresee the recovery trend after the financial crisis in 2008, and another round of Alasso further worsens TAlasso. As shown in [Fig. 2](#), when $h = 1$, Plasso's forecast coincides with the movement of the realization during the recovery period after 2008, whereas Alasso/TAlasso moves in the opposite direction. Whereas large deviation exacerbates RMPSE, under MPAE, the gap between Plasso and Alasso/TAlasso in this case is narrowed or even reversed. Under the 10-year rolling window, all of the methods encounter difficulty around the financial crisis, and the difference between TAlasso and Slaslo is negligible. Thus, we view the unsatisfactory RMPSE of Alasso/TAlasso here as an adverse case under the specific rolling window.

In terms of prediction performance, it is known that eliminating irrelevant predictors is more important than including relevant predictors; see, for example, [Ploberger and Phillips \(2003\)](#). The inclusion of the irrelevant predictors could be detrimental in forecasting contexts, and including irrelevant nonstationary predictors in predictive regression can be harmful because stock returns are stationary. In this sense, Alasso and TAlasso provide more conservative variable selection in predictive regressions. In [Fig. 3](#), the instance with $h = 1/2$ and the 15-year rolling window is used to illustrate

² BIC is a popular choice geared toward variable selection, but in our context, it is incompatible with the Plasso/Slaslo, which cannot cope with variable screening and consistent estimation simultaneously.

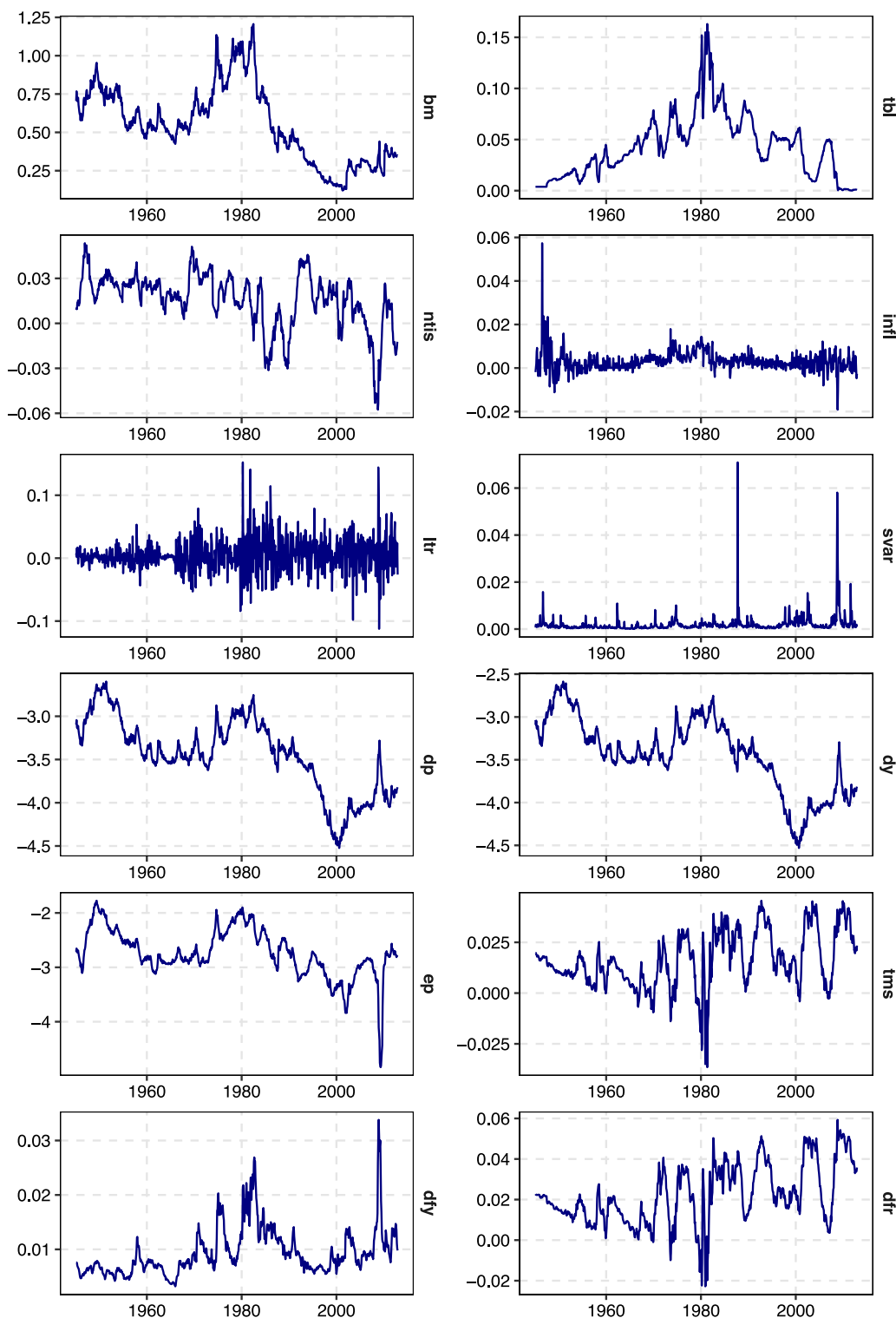


Fig. 1. Time plots of the 12 predictors from the dataset of Welch and Goyal (2008). Note: The persistent regressors include the dividend price ratio (*dp*), dividend yield (*dy*), earning price ratio (*ep*), term spread (*tms*), default yield spread (*dfy*), default return spread (*dfr*), book-to-market ratio (*bm*), net equity expansion (*ntis*), and treasury bill rates (*tbl*), whose estimated AR(1) coefficients are greater than 0.95. The stationary predictors are the long-term return of government bonds (*ltr*), stock variance (*svar*), and inflation (*infl*).

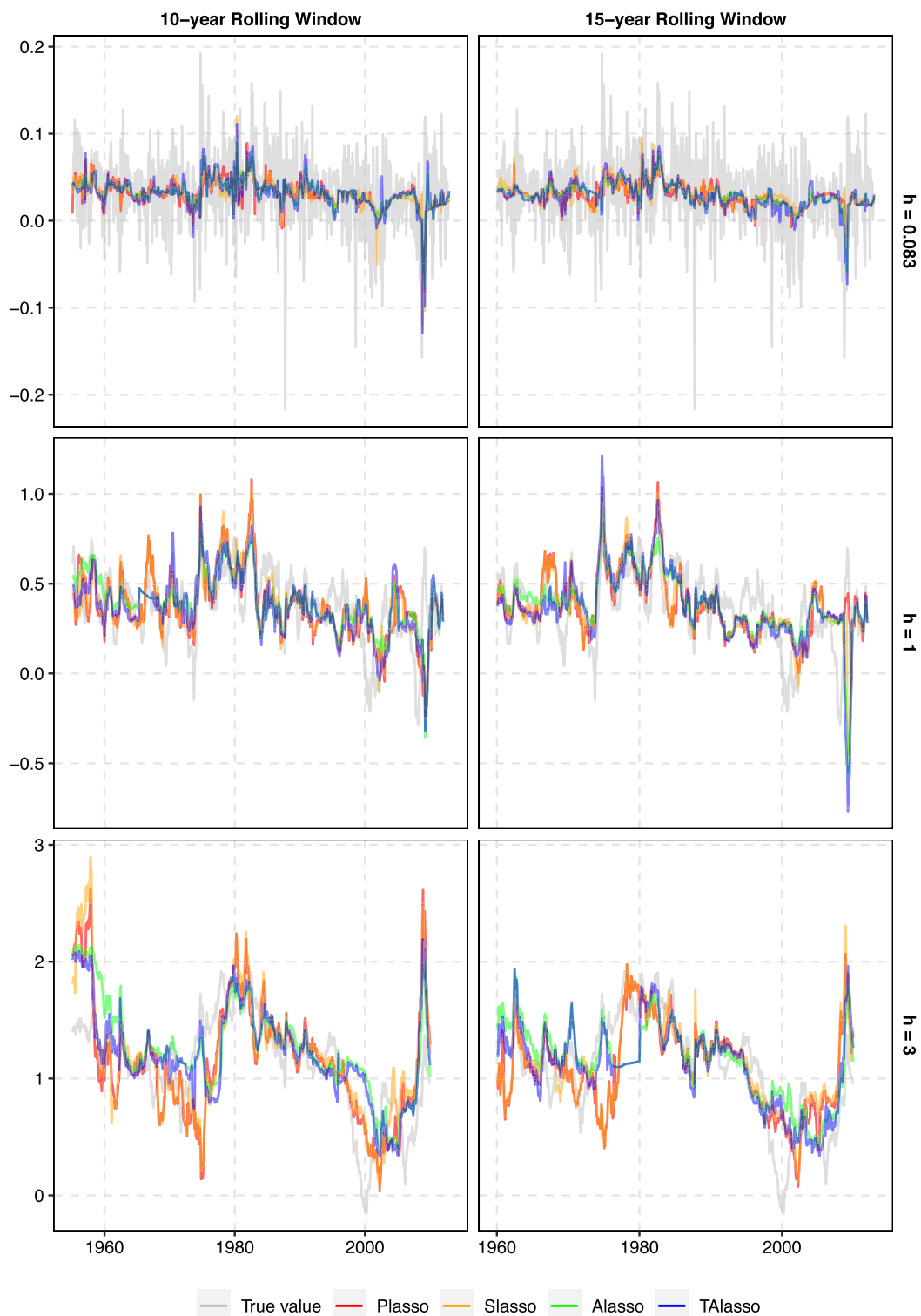


Fig. 2. Realized return versus predicted returns. h is selected as $1/12$, 1, and 3 to present the short, medium, and long horizons, respectively.

the estimated coefficients under CV. The shrinkage methods select different variables over the estimation windows, indicating the evolution of the predictive models across time. Alasso and TAlasso throw out more variables than Plasso

Table 4
RMPSE and MPAA for predicting S&P 500 excess return.

Tuning para.		OLS NA	RWwD NA	Plas. CV	Slas. CV	Alas. CV	TAlas. CV	Alas. BIC	TAlas. BIC
	<i>h</i>	RMPSE $\times$ 100							
10-year rolling window	1/12	1.000	0.950	0.944	0.947	0.932	0.952	0.949	0.950
	1/4	1.000	0.842	0.945	0.900	0.806	0.821	0.835	0.839
	1/2	1.000	0.946	0.937	0.914	0.862	0.883	0.919	0.916
	1	1.000	1.123	0.953	0.933	0.958	0.947	0.943	0.912
	2	1.000	1.297	0.962	0.904	0.898	0.905	0.893	0.865
	3	1.000	1.173	0.892	0.941	0.854	0.806	0.855	0.774
15-year rolling window	1/12	1.000	0.982	0.956	0.960	0.952	0.963	0.980	0.984
	1/4	1.000	0.915	0.891	0.889	0.871	0.874	0.921	0.924
	1/2	1.000	0.924	0.958	0.919	0.882	0.917	0.903	0.918
	1	1.000	1.055	0.859	0.961	0.917	1.029	0.934	0.962
	2	1.000	1.518	0.940	0.952	0.860	0.818	0.865	0.820
	3	1.000	1.617	0.974	1.002	0.953	0.890	0.843	0.824
	<i>h</i>	MPAA $\times$ 100							
10-year rolling window	1/12	1.000	0.944	0.948	0.938	0.928	0.951	0.942	0.947
	1/4	1.000	0.925	0.933	0.907	0.862	0.876	0.903	0.920
	1/2	1.000	0.955	0.922	0.909	0.851	0.863	0.915	0.917
	1	1.000	1.140	0.957	0.944	0.929	0.939	0.925	0.895
	2	1.000	1.356	0.903	0.886	0.824	0.834	0.831	0.807
	3	1.000	1.391	0.940	0.960	0.912	0.836	0.896	0.814
15-year rolling window	1/12	1.000	0.979	0.971	0.969	0.970	0.984	0.979	0.989
	1/4	1.000	0.932	0.905	0.906	0.875	0.884	0.926	0.925
	1/2	1.000	0.983	0.962	0.942	0.893	0.934	0.903	0.928
	1	1.000	1.107	0.921	0.970	0.853	0.933	0.879	0.878
	2	1.000	1.705	0.916	0.893	0.812	0.805	0.804	0.777
	3	1.000	1.907	1.010	1.015	1.040	0.966	0.897	0.882

Note: The upper panel compares the root mean prediction squared error (RMPSE), defined as $\sqrt{E[(y_n - \hat{y}_n)^2]}$, and the lower panel compares the mean prediction absolute error $E[|y_n - \hat{y}_n|]$. The RMPSE and MPAA of OLS are normalized to be 1. Bold numbers indicate the best performance in each row, and italic numbers indicate the worst.

or Lasso and hence deliver more parsimonious models. For example, they completely eliminate the variables *ltr* and *infl*.

To highlight the potentially cointegrated variables, Fig. 4(a) presents the Alasso and TAllasso estimates of (*dp*, *dy*) and (*dfr*, *tms*) by CV, and in Fig. 4(b), BIC decides the tuning parameters and then produces the estimates. The table ahead of the subfigures lists the fraction of the active Alasso estimates annihilated by TAllasso over the rolling windows. In more than a quarter of the cases when Alasso selects *dy* or *tms*, TAllasso further penalizes the corresponding coefficient to be 0 in the second round. In general, BIC tends to freeze out more variables than CV, especially for *dy* and *tms*. Although the numbers vary, similar patterns are found in other combinations of forecast horizons and rolling window lengths.

7. Conclusion

We explore LASSO procedures in the presence of stationary, nonstationary, and cointegrated predictors. Although it does not enjoy the well-known oracle property, Alasso breaks the links between inactive cointegration groups, and its repeated implementation TAllasso recovers the oracle property because of the differentiated penalty on the zero and nonzero coefficients. TAllasso is adaptive to a system of multiple predictors with various degrees of persistence, unlike Plasso's uniform penalty or Slasso's penalty based only on the marginal variation of each predictor. Moreover, TAllasso saves the effort of sorting the predictors according to their degrees of persistence, so we can be agnostic regarding the time series properties of the predictors. The automatic penalty adjustment of TAllasso guarantees consistent model selection and the optimal rate of convergence. Such desirable properties may improve the out-of-sample prediction in practice under complex predictive environments with a mixture of regressors.

To focus on the mixed root setting, we adopt the simplest asymptotic framework with a fixed p and $n \rightarrow \infty$ to demonstrate the clear contrast between OLS, Alasso, TAllasso, Plasso, and Slasso. This asymptotic framework is in line with the state of the art of the predictive regression studies in financial econometrics (Kostakis et al., 2014; Phillips and Lee, 2016; Xu, 2018). However, the large number of potential regressors available in the era of big data indicate the need for a theoretical extension to allow for an infinite number of regressors in the limit. As the restricted eigenvalue condition (Bickel et al., 2009) is unsuitable in our context, in which the nonstationary part of the Gram matrix does not degenerate, a few papers (Onatski and Wang, 2018, 2020; Zhang et al., 2019; Bykhovskaya and Gorin, 2020) point to the spectral analysis of a large random matrix (Bai and Silverstein, 2010) as the foundation of asymptotic analysis. In future

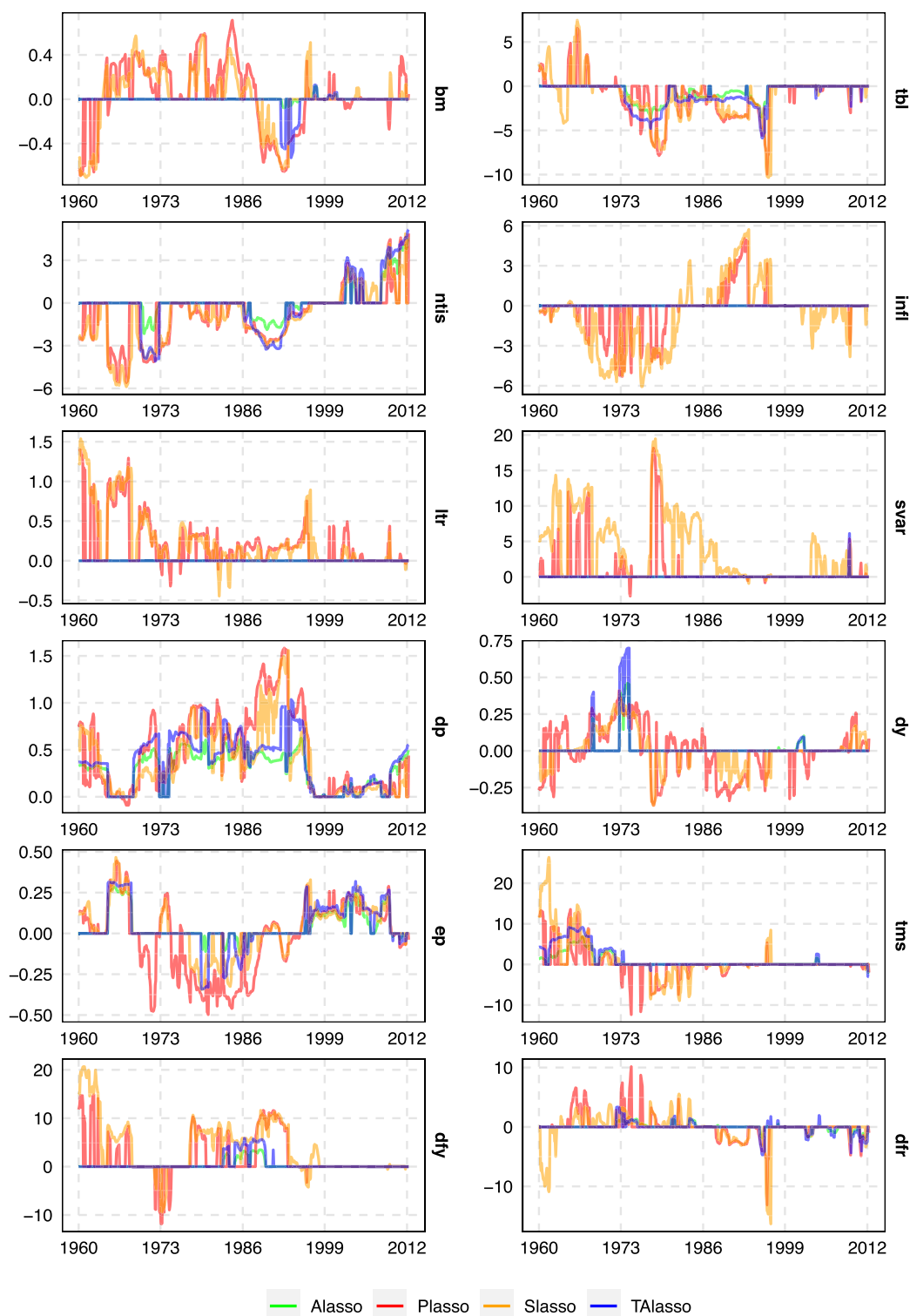
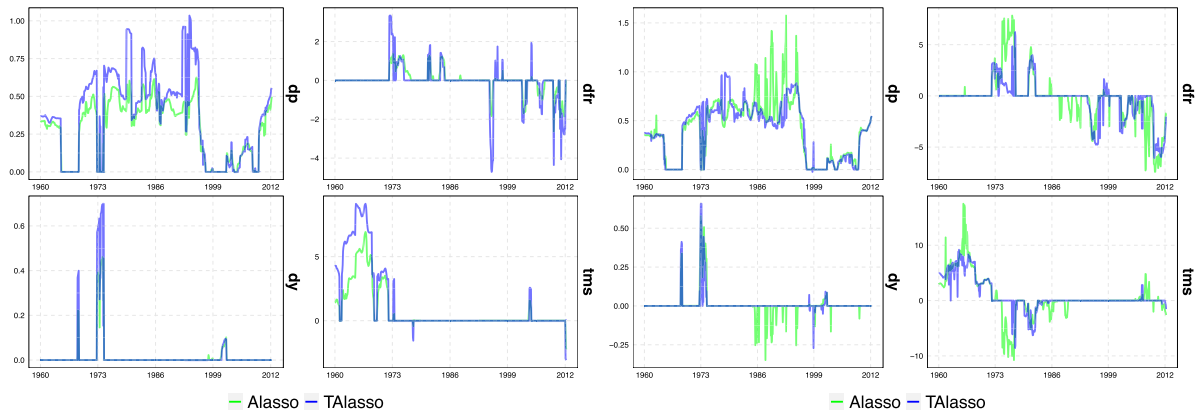


Fig. 3. Estimated coefficients for all 12 variables (15-year rolling window, $h = 1/2$).

research, we will apply the new technical apparatus to deal with the minimal eigenvalue of the Gram matrix to generalize the insight gleaned from low-dimensional asymptotics to high-dimensional asymptotics.

Fraction of active Alasso estimates eliminated by TAlasso

	dp	dy	dfr	tms
CV	0.018	0.250	0.263	0.014
BIC	0.016	0.460	0.236	0.308



(4.a) CV

(4.b) BIC

Fig. 4. Estimated coefficient of potentially cointegrated variables (15-year rolling window, $h = 1/2$).

Another line of related literature concerns uniformly valid inference and forecasting after LASSO model selection; see [Belloni et al. \(2018b,a\)](#) or [Hirano and Wright \(2017\)](#), for example. These papers allow for model selection error by LASSO and provide valid inference or prediction by introducing local limit theory with only small departures from the true models. Combining these recent developments with our current LASSO theory is another exciting direction for future research.

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Appendix. Technical appendix

A.1. Proofs in Section 2

Proof of Theorem 1. Since only Alasso is considered in this proof, we denote $\hat{\beta} = \hat{\beta}^A$ and $\hat{M} = \hat{M}^A$ for conciseness. Let $\beta_n = \beta_n^* + n^{-1}v$ be a perturbation around the true parameter β_n^* , and let

$$\Psi_n(v) = \|Y - \sum_{j=1}^p x_j(\beta_{jn}^* + \frac{v_j}{n})\|^2 + \lambda_n \sum_{j=1}^p \hat{\tau}_j |\beta_{jn}^* + \frac{v_j}{n}|.$$

Define $\hat{v}^{(n)} = n(\hat{\beta} - \beta_n^*)$. The fact that $\hat{\beta}$ minimizes (6) implies $\hat{v}^{(n)} = \arg \min_v \Psi_n(v)$. Let

$$\begin{aligned} V_n(v) &= \Psi_n(v) - \Psi_n(0) = \|u - \frac{X'v}{n}\|^2 - \|u\|^2 + \lambda_n \left(\sum_{j=1}^p \hat{\tau}_j |\beta_{jn}^* + \frac{v_j}{n}| - \sum_{j=1}^p \hat{\tau}_j |\beta_{jn}^*| \right) \\ &= v' \left(\frac{X'X}{n^2} \right) v - 2 \frac{u'X}{n} v + \lambda_n \sum_{j=1}^p \hat{\tau}_j (|\beta_{jn}^* + \frac{v_j}{n}| - |\beta_{jn}^*|). \end{aligned} \quad (\text{A.1})$$

The first and the second terms on the right-hand side of (A.1) converge in distribution, as $n^{-2}X'X \Rightarrow \Omega$ and $n^{-1}X'u = n^{-1} \sum_{i=1}^n x'_i u_i \Rightarrow \zeta$, by the functional central limit theorem (FCLT) and the continuous mapping theorem. The third term involves the weight $\hat{\tau}_j = |\hat{\beta}_j^{\text{ols}}|^{-\gamma}$ for each j . Since the OLS estimator $n(\hat{\beta}^{\text{ols}} - \beta_n^*) \Rightarrow \Omega^{-1}\zeta = O_p(1)$, we have

$$\hat{\tau}_j = |\beta_{jn}^* + O_p(n^{-1})|^{-\gamma} = |\beta_{jn}^{0*}/n^{\delta_j} + O_p(n^{-1})|^{-\gamma}. \quad (\text{A.2})$$

If $\beta_{jn}^{0*} \neq 0$, then β_{jn}^* dominates $n^{-1}v_j$ for a large n and

$$|\beta_{jn}^* + \frac{v_j}{n}| - |\beta_{jn}^*| = n^{-1}v_j \text{sgn}(\beta_{jn}^*) = n^{-1}v_j \text{sgn}(\beta_{jn}^{0*}). \quad (\text{A.3})$$

Now (A.2) and (A.3) imply

$$\begin{aligned} \lambda_n \hat{\tau}_j \cdot (|\beta_{jn}^* + \frac{v_j}{n}| - |\beta_{jn}^*|) &= \frac{\lambda_n}{n|\beta_{jn}^{0*}/n^{\delta_j} + O_p(n^{-1})|^\gamma} v_j \text{sgn}(\beta_{jn}^{0*}) = \frac{\lambda_n n^{\delta_j \gamma - 1}}{|\beta_{jn}^{0*} + o_p(1)|^\gamma} v_j \text{sgn}(\beta_{jn}^{0*}) \\ &= O_p(\lambda_n n^{\delta_j \gamma - 1}) = o_p(1) \end{aligned} \quad (\text{A.4})$$

by the given rate of λ_n . On the other hand, if $\beta_{jn}^{0*} = 0$, then $(|\beta_{jn}^* + n^{-1}v_j| - |\beta_{jn}^*|) = n^{-1}|v_j|$. For any fixed $v_j \neq 0$,

$$\lambda_n \hat{\tau}_j \cdot (|\beta_{jn}^* + \frac{v_j}{n}| - |\beta_{jn}^*|) = \frac{\lambda_n}{n|\hat{\beta}_j^{\text{ols}}|^\gamma} |v_j| = \frac{\lambda_n n^{\gamma-1}}{n|\hat{\beta}_j^{\text{ols}}|^\gamma} |v_j| = \frac{\lambda_n n^{\gamma-1}}{O_p(1)} |v_j| \rightarrow \infty \quad (\text{A.5})$$

since $\lambda_n \rightarrow \infty$, $\gamma \geq 1$ and the OLS estimator is asymptotically non-degenerate. Thus we have $V_n(v) \Rightarrow V(v)$ for every fixed v , where

$$V(v) = \begin{cases} v' \Omega v - 2v' \zeta, & \text{if } v_{M^{*c}} = \mathbf{0}_{|M^{*c}|} \\ \infty, & \text{otherwise.} \end{cases}$$

Both $V_n(v)$ and $V(v)$ are strictly convex in v , and $V(v)$ is uniquely minimized at

$$\begin{pmatrix} v_{M^*} \\ v_{M^{*c}} \end{pmatrix} = \begin{pmatrix} \Omega_{M^*}^{-1} \zeta_{M^*} \\ \mathbf{0} \end{pmatrix}.$$

Applying the Convexity Lemma (Pollard, 1991), we have

$$\hat{v}_{M^*}^{(n)} = n(\hat{\beta}_{M^*} - \beta_{M^*}^*) \Rightarrow \Omega_{M^*}^{-1} \zeta_{M^*} \quad \text{and} \quad \hat{v}_{M^{*c}}^{(n)} \Rightarrow \mathbf{0}. \quad (\text{A.6})$$

The first part of the above result establishes Theorem 1(b) about the asymptotic distribution for the coefficients in M^* .

Next, we show variable selection consistency. The result $P(M^* \subseteq \hat{M}) \rightarrow 1$ immediately follows by the first part of (A.6), since $\hat{v}_{M^*}^{(n)}$ converges in distribution to a non-degenerate continuous random variable. For those $j \in M^{*c}$, if the event $\{j \in \hat{M}\}$ occurs, then the KKT condition entails

$$\frac{1}{n} x'_j (y - X \hat{\beta}) = \text{sgn}(\hat{\beta}_j) \frac{\lambda_n \hat{\tau}_j}{2n}. \quad (\text{A.7})$$

Notice that the order on the right-hand side is governed by

$$\frac{\lambda_n \hat{\tau}_j}{n} = \frac{\lambda_n}{n|\hat{\beta}_j^{\text{ols}}|^\gamma} = \frac{\lambda_n n^{\gamma-1}}{n|\hat{\beta}_j^{\text{ols}}|^\gamma} = \frac{\lambda_n n^{\gamma-1}}{O_p(1)} \rightarrow \infty \quad (\text{A.8})$$

given the rate of λ_n . However, using $y = X\beta_n^* + u$ and (A.6), the left-hand side of (A.7) is

$$\begin{aligned} \frac{1}{n} x'_j (y - X \hat{\beta}) &= \frac{1}{n} x'_j (X\beta_n^* - X \hat{\beta} + u) = \frac{x'_j X}{n^2} \cdot n(\beta_n^* - \hat{\beta}) + \frac{x'_j u}{n} \\ &= -\frac{x'_j}{n^2} (X_{M^*} \hat{v}_{M^*}^{(n)} + X_{M^{*c}} \hat{v}_{M^{*c}}^{(n)}) + \frac{x'_j u}{n} \\ &\Rightarrow -\Omega_{j,M^*} \Omega_{M^*}^{-1} \zeta_{M^*} - \Omega_{j,M^{*c}} o_p(1) + \zeta_j = O_p(1), \end{aligned} \quad (\text{A.9})$$

where $\Omega_{j,M^*} = [\Omega_{j,l}]_{l \in M^*}$ and $\Omega_{j,M^{*c}}$ is defined similarly. In other words, the left-hand side of (A.7) remains a non-degenerate continuous random variable in the limit. For any $j \in M^{*c}$, the disparity of the two sides of the KKT condition implies

$$P(j \in \hat{M}_n) = P\left(\frac{1}{n} x'_j (y - X \hat{\beta}) = \text{sgn}(\hat{\beta}_j) \frac{\lambda_n \hat{\tau}_j}{2n}\right) \rightarrow 0.$$

That is, $P(M^{*c} \subseteq \hat{M}) \rightarrow 0$ or equivalently $P(\hat{M} \subseteq M^*) \rightarrow 1$. We thus conclude the variable selection consistency in Theorem 1(a) ■

A.2. Proofs in Section 3

To express the OLS asymptotic distribution, we need some extra notations. Define $\Omega = \sum_{h=-\infty}^{\infty} \mathbb{E}(\xi_i \xi_{i-h}')$ as the long-run covariance matrix associated with the innovation vector. It can be written as is

$$\Omega_{(p+1) \times (p+1)} = F(1) \Sigma_{\varepsilon} F(1)' = \begin{pmatrix} \Omega_{zz} & \Omega_{zv} & \Omega_{ze} & 0 \\ \Omega'_{zv} & \Omega_{vv} & \Omega_{ve} & 0 \\ \Omega'_{ze} & \Omega'_{ve} & \Omega_{ee} & \Omega_{eu} \\ 0 & 0 & \Omega'_{eu} & \Omega_{uu} \end{pmatrix}$$

according to the explicit form of Σ_{ε} , where $F(1) = (F'_z(1), F'_v(1), F'_e(1), F'_u(1))'$. Moreover, define the sum of one-sided autocovariance as $\Lambda = \sum_{h=1}^{\infty} \mathbb{E}(\xi_i \xi_{i-h}')$, and $\Delta = \Lambda + \mathbb{E}(\xi_i \xi_i')$. We use the functional law (Phillips and Solo, 1992) under [Assumption 2](#) to derive

$$\frac{1}{\sqrt{n}} \sum_{i=1}^{\lfloor nr \rfloor} \xi_i = (B'_{zn}(r), B'_{vn}(r), B'_{en}(r), B_u(r))' \\ \implies (B'_z(r), B'_v(r), B'_e(r), B_u(r))' := B_{\xi}(r) \sim BM(\Omega),$$

where $B_{\xi}(r)$ is a vector Brownian motion whose covariance kernel is Ω . Define the corresponding vector Ornstein Uhlenbeck process $J_{c_2}(r)$ associated with $B_{v_2}(r)$, which is a subvector of $B_v(r) = (B'_{v_1}(r), B'_{v_2}(r))'$ in the standard way:

$$dJ_{c_2}(r) = c_2 J_{c_2}(r) dr + dB_{v_2}(r), \quad J_{c_2}(0) = 0.$$

Similarly, define $J_{c_x}(r)$ as $dJ_{c_x}(r) = c_x J_{c_x}(r) dr + dB_e(r)$, $J_{c_x}(0) = 0$.

Lemma 1. If the linear model (10) satisfies [Assumption 2](#), then $(u'Z^+/\sqrt{n}, u'X^+/n)$ converges in distribution to a non-degenerate stable law.

Proof of Lemma 1. The left-top $p \times p$ submatrix of Ω can be represented in a conformable manner as

$$\begin{pmatrix} \Omega_{zz} & \Omega_{zv} & \Omega_{ze} \\ \Omega'_{zv} & \Omega_{vv} & \Omega_{ve} \\ \Omega'_{ze} & \Omega'_{ve} & \Omega_{ee} \end{pmatrix} = \begin{pmatrix} \Omega_{zz}^+ & \Omega_{zx}^+ \\ \Omega_{zx}^+{}' & \Omega_{xx}^+ \end{pmatrix} \begin{matrix} (p_z+1) \times (p_z+p_1) \\ (p_2+p_x) \times (p_2+p_x) \end{matrix}. \quad (\text{A.10})$$

The Beveridge–Nelson (BN) decomposition and central limit theorem lead to

$$Z^+u/\sqrt{n} \implies \zeta_{z^+} \sim N(0, \Sigma_{uu}\Omega_{zz}^+). \quad (\text{A.11})$$

The columns in X^+ are local unit root processes with no cointegration relationship. Let the i th row of X^+ be $X_i^+ = (X_{2i}^c, X_i)^'$, a (p_2+p_x) -by-1 vector. Using the component-wise BN decomposition, the scalar $u_i = \varepsilon_i' \times F_u(1) - \Delta \tilde{\varepsilon}_{ui}$, where $\Delta \tilde{\varepsilon}_{ui}$ is the last $(p+1)$ -th scalar component of the vector $\Delta \tilde{\varepsilon}_i = \tilde{\varepsilon}_i - \tilde{\varepsilon}_{i-1}$ and $\tilde{\varepsilon}_i = \sum_{j=0}^{\infty} \tilde{F}_j \varepsilon_{i-j}$ with $\tilde{F}_j = \sum_{k=j+1}^{\infty} F_k$. Thus we have

$$\frac{1}{n} X^{+u} = \frac{1}{n} \sum_{i=1}^n X_i^+ u_i = \left(\frac{1}{n} \sum_{i=1}^n X_i^+ \varepsilon_i' \right) F_u(1) - \frac{1}{n} \sum_{i=1}^n X_i^+ \Delta \tilde{\varepsilon}_{ui}.$$

Functional CLT combined with the stationary $O_p(1)$ initial condition provides $X_i^+/\sqrt{n} \implies J_{x^+}(r) = (J'_{c_2}(r), J'_{c_x}(r))'$, and thus $n^{-1} \sum_{i=1}^n X_i^+ \varepsilon_i' \implies \int_0^1 J_{x^+}(r) dB_{\varepsilon}(r)'$. The summation by parts now implies

$$\frac{1}{n} \sum_{i=1}^n X_i^+ \Delta \tilde{\varepsilon}_{ui} = -\frac{1}{n} \sum_{i=1}^n u_{xi}^+ \tilde{\varepsilon}_{ui-1} + o_p(1) \xrightarrow{p} \Delta_{+u}$$

where $\Delta_{+u} = \sum_{h=0}^{\infty} E[\tilde{u}_i u_{i-h}]$ is the corresponding submatrix of the one-sided long-run covariance and $u_{xi}^+ = X_i^+ - X_{i-1}^+$. Combining these results, we have

$$\frac{X^{+u}}{n} \implies \zeta_{x^+} \sim \int_0^1 J_{x^+}(r) dB_{\varepsilon}(r)' F_u(1) + \Delta_{+u}. \quad (\text{A.12})$$

Since the marginal limit distributions are non-degenerate in (A.11) and (A.12), so does the joint limiting distribution. ■

Proof of Theorem 2. We transform and scale-normalize the OLS estimator as

$$R_n Q (\hat{\theta}^{\text{ols}} - \theta_n^*) = R_n Q (W'W)^{-1} W'u \\ = R_n Q (W'W)^{-1} Q'R_n (Q'R_n)^{-1} W'u \\ = [R_n^{-1} Q'^{-1} W'WQ^{-1} R_n^{-1}]^{-1} R_n^{-1} Q'^{-1} W'u. \quad (\text{A.13})$$

The first factor

$$R_n^{-1} Q'^{-1} W' W Q^{-1} R_n^{-1} = \begin{pmatrix} \frac{Z^{++} Z^+}{n} & \frac{Z^{++} X^+}{n^{3/2}} \\ \frac{Z^{++} X^+}{n^{3/2}} & \frac{X^{++} X^+}{n^2} \end{pmatrix} \Rightarrow \begin{pmatrix} \Omega_{zz}^+ & 0 \\ 0 & \int_0^1 J_{x^+}(r) J_{x^+}(r)' dr \end{pmatrix} =: \Omega^+ \quad (\text{A.14})$$

where Ω_{zz}^+ is defined in (A.10), and $J_{x^+}(r) = (J'_{c_2}(r), J'_{c_x}(r))'$. The second factor

$$R_n^{-1} Q'^{-1} W' u = \begin{pmatrix} Z^{++} u / \sqrt{n} \\ X^{++} u / n \end{pmatrix} \Rightarrow \zeta^+, \quad (\text{A.15})$$

where the marginal distributions of ζ^+ can be found in (A.11) and (A.12). Thus the stated conclusion follows by the continuous mapping theorem. ■

Remark 8. In the rotated coordinate system, (13) in Theorem 2 and the definition of ζ_{x^+} imply that an asymptotic bias term Δ_{+u} appears in the limit distribution of OLS with nonstationary predictors. This asymptotic bias arises from the serial dependence in the innovations. However, the asymptotic bias does not affect the rate of convergence as $Q(\hat{\theta}^{\text{ols}} - \theta_n^*) = O_p(\text{diag}(R_n^{-1}))$.

Remark 9. The OLS estimator will serve as the initial estimator for Alasso. In principle either the fully-modified OLS (Phillips and Hansen, 1990) or the canonical cointegrating regression estimator (Park, 1992) can be an initial estimator as well, because their rates of convergence are the same as those of OLS.

To simplify notation, define R_{jn} as the j th diagonal element of R_n , $\hat{C} = \hat{M}^A \cap \mathcal{C}$ and $C^* = M^* \cap \mathcal{C}$. Recall $C^{*c} = M^{*c} \cap \mathcal{C}$.

Proof of Theorem 3. Again, we use $\hat{\theta} = \hat{\theta}^A$ and $\hat{M} = \hat{M}^A$ for conciseness in this proof. For a constant nonzero vector

$$\tilde{v} = (\tilde{v}'_z, \tilde{v}'_1, \mathbf{0}'_{p_2}, \tilde{v}'_x)' \neq \mathbf{0} \quad (\text{A.16})$$

where the elements associated with c_2 are suppressed as $\mathbf{0}$, we add a local perturbation

$$v_n = Q^{-1} R_n^{-1} \tilde{v} = \begin{pmatrix} \frac{\tilde{v}'_z}{\sqrt{n}}, \frac{\tilde{v}'_1}{\sqrt{n}}, -\frac{\tilde{v}'_1 A_1}{\sqrt{n}}, \frac{\tilde{v}'_x}{n} \end{pmatrix}$$

to θ_n^* so that the perturbed coefficient $\theta_n = \theta_n^* + v_n$. Let

$$\Psi_n(\tilde{v}) = \|Y - W(\theta_n^* + v_n)\|^2 + \lambda_n \sum_{j=1}^p \hat{\tau}_j |\theta_{jn}^* + v_{jn}|,$$

and then define

$$\begin{aligned} V_n(\tilde{v}) &= \Psi_n(\tilde{v}) - \Psi_n(\mathbf{0}) = \|u - W v_n\|^2 - \|u\|^2 + \lambda_n \sum_{j=1}^p \hat{\tau}_j (|\theta_{jn}^* + v_{jn}| - |\theta_{jn}^*|) \\ &= v_n' W' W v_n - 2v_n' W' u + \lambda_n \sum_{j=1}^p \hat{\tau}_j (|\theta_{jn}^* + v_{jn}| - |\theta_{jn}^*|). \end{aligned}$$

We have shown in the proof of Theorem 2 that the first term

$$v_n' W' W v_n = \tilde{v}' R_n^{-1} Q'^{-1} W'^{-1} R_n^{-1} \tilde{v} \Rightarrow \tilde{v}' \Omega^+ \tilde{v} \quad (\text{A.17})$$

by (A.14), and the second term

$$2v_n' W' u = 2\tilde{v}' R_n^{-1} Q'^{-1} W' u \Rightarrow 2\tilde{v}' \zeta^+ \quad (\text{A.18})$$

by (A.15).

We focus on the third term. Theorem 2 has shown that the OLS estimator $\hat{\theta}_j^{\text{ols}} - \theta_j^* = O_p(R_{jn}^{-1})$ for each $j \in \mathcal{M}_Q$. Given any fixed $\tilde{v}_j \neq 0$ and a sufficiently large n :

- For $j \in \mathcal{I}_0 \cup \mathcal{C}_1$, by the definition of v each element $v_{jn} = \tilde{v}_j / \sqrt{n}$. If $\theta_j^{0*} \neq 0$, we have $|\theta_j^* + \tilde{v}_j / \sqrt{n}| - |\theta_j^*| = n^{-1/2} \tilde{v}_j \text{sgn}(\theta_j^{0*})$, and thus $\lambda_n \hat{\tau}_j (|\theta_j^* + \tilde{v}_j / \sqrt{n}| - |\theta_j^{0*}|) = O_p(\lambda_n n^{-1/2}) = o_p(1)$. If $\theta_j^{0*} = 0$, we have $\lambda_n \hat{\tau}_j (|\theta_j^* + \tilde{v}_j / \sqrt{n}| - |\theta_j^{0*}|) = \lambda_n n^{0.5(\gamma-1)} |\tilde{v}_j| / O_p(1) = O_p(\lambda_n n^{(\gamma-1)/2}) \rightarrow \infty$ when $\tilde{v}_j \neq 0$.
- For $j \in \mathcal{I}_1$, the true coefficient $\theta_{jn}^* = \theta_j^{0*} / n^{\delta_j}$ depends on n while the perturbation $v_{jn} = \tilde{v}_j / n$. If $\theta_j^{0*} \neq 0$, then θ_{jn}^* dominates $\tilde{v}_j / n$ in the limit and $(|\theta_{jn}^* + \tilde{v}_j / n| - |\theta_{jn}^*|) = \text{sgn}(\theta_j^{0*}) \tilde{v}_j / n$. By the same derivation in (A.4), $\lambda_n \hat{\tau}_j (|\theta_{jn}^* + \tilde{v}_j / n| - |\theta_{jn}^*|) = O_p(\lambda_n n^{(\delta_j \gamma - 1)}) = o_p(1)$ given the condition (17). If $\theta_j^{0*} = 0$, according to the derivation in (A.5) $\lambda_n \hat{\tau}_j (|\theta_{jn}^* + \tilde{v}_j / n| - |\theta_{jn}^*|) = \lambda_n n^{\gamma-1} |\tilde{v}_j| / O_p(1) = O_p(\lambda_n n^{\gamma-1}) \rightarrow \infty$ when $\tilde{v}_j \neq 0$.

The above analysis indicates $V_n(\tilde{v}) \implies V(\tilde{v})$ for every fixed $\tilde{v}$ in (A.16), where

$$V(\tilde{v}) = \begin{cases} \tilde{v}'\Omega^+\tilde{v} - 2\tilde{v}'\zeta^+, & \text{if } \tilde{v}_{M_Q^{*c}} = 0 \\ \infty, & \text{otherwise.} \end{cases}$$

Let $\hat{v}^{(n)} = \hat{\theta} - \theta_n^*$. The same argument about the strict convexity of $V_n(\tilde{v})$ and $V(\tilde{v})$ implies

$$(R_n Q \hat{v}^{(n)})_{M_Q^*} \implies (\Omega_{M_Q^*}^+)^{-1} \zeta_{M_Q^*}^+ \quad (\text{A.19})$$

$$(R_n Q \hat{v}^{(n)})_{M_Q^{*c}} \implies 0 \quad (\text{A.20})$$

We have established Theorem 3(a).

Next, we move on to discuss the effect of variable selection. Consider any $j \in \hat{M}$. The KKT condition with respect to θ_j entails

$$W_j'(y - W\hat{\theta}) = \text{sgn}(\hat{\theta}_j) \lambda_n \hat{\tau}_j / 2. \quad (\text{A.21})$$

We will invoke similar argument as in (A.8) and (A.9) to show the disparity of the two sides of the KKT condition. The left-hand side of (A.21) is the j th element of the $p \times 1$ vector $W'(y - W\hat{\theta})$. Pre-multiply the diagonal matrix $R_n^{-1}Q'^{-1}$ to the vector:

$$\begin{aligned} R_n^{-1}Q'^{-1}W'(y - W\hat{\theta}) &= R_n^{-1}Q'^{-1}W' \left(W(\theta_n^* - \hat{\theta}) + u \right) \\ &= (R_n^{-1}Q'^{-1}W'WQ^{-1}R_n^{-1}) R_n Q(\theta_n^* - \hat{\theta}) + R_n^{-1}Q'^{-1}W'u \\ &= R_n^{-1}Q'^{-1}W'WQ^{-1}R_n^{-1}O_p(1) + R_n^{-1}Q'^{-1}W'u \\ &= (\Omega^+ + o_p(1)) O_p(1) + (\zeta^+ + o_p(1)) = O_p(1) \end{aligned} \quad (\text{A.22})$$

where the third equality follows by (A.19) and (A.20), and the fourth equality by (A.14) and (A.15). Notice that (A.22) gives the biggest order of the left-hand side, while some components of this $p \times 1$ vector may be degenerate (of order $o_p(1)$).

Suppose $j \in M^{*c}$. For $j \in \mathcal{I}$, the rotation Q does not change these variables so the order of the left-hand side of (A.21) is the same as (A.22). If $j \in \mathcal{I}_0$, multiply $n^{-1/2}$ to the right-hand side of (A.21):

$$\frac{\text{sgn}(\hat{\theta}_j)}{2\sqrt{n}} \lambda_n \hat{\tau}_j = \text{sgn}(\hat{\theta}_j) \frac{\lambda_n}{2\sqrt{n}|\hat{\theta}_j^{\text{ols}}|^\gamma} = \text{sgn}(\hat{\theta}_j) \frac{\lambda_n n^{0.5(\gamma-1)}}{2|\sqrt{n}\hat{\theta}_j^{\text{ols}}|^\gamma} = \text{sgn}(\hat{\theta}_j) O_p(\lambda_n n^{0.5(\gamma-1)}) \rightarrow \infty \text{ or } -\infty$$

as $\gamma \geq 1$ and $\lambda_n \rightarrow \infty$. Similarly, if $j \in \mathcal{I}_1$ we multiply n^{-1} to the right-hand side of (A.21):

$$\frac{\text{sgn}(\hat{\theta}_j)}{2n} \lambda_n \hat{\tau}_j = \text{sgn}(\hat{\theta}_j) \frac{\lambda_n n^{(\gamma-1)}}{2|n\hat{\theta}_j^{\text{ols}}|^\gamma} = \text{sgn}(\hat{\theta}_j) O_p(\lambda_n n^{(\gamma-1)}) \rightarrow \infty \text{ or } -\infty.$$

We have verified that for $j \in \mathcal{I}$ the right-hand side of (A.21) is of bigger order than its left-hand side. It immediately follows that given the specified rate of λ_n , for any $j \in M^{*c} \cap \mathcal{I}$ we have (20) since $P(j \in \hat{M} \cap M^{*c} \cap \mathcal{I}) \rightarrow P(O_p(1) = \infty \text{ or } -\infty) = 0$. We have established (20).

Estimation consistency (18) immediately implies (21). On the other hand, it is possible that variables in $\mathcal{C}$ are wrongly selected into $\hat{M}$. If the event $\{j \in \hat{M}\}$ occurs for some $j \in \mathcal{C}^{*c}$, the KKT condition may still hold in the limit. To see this, pre-multiply n^{-1} to the left-hand side of (A.21) and it is of order $O_p(1)$ according to (A.22), and it can be degenerate. On the other hand, the right-hand side becomes

$$\frac{\text{sgn}(\hat{\theta}_j)}{2n} \lambda_n \hat{\tau}_j = \text{sgn}(\hat{\theta}_j) \frac{\lambda_n n^{(0.5\gamma-1)}}{2|\sqrt{n}\hat{\theta}_j^{\text{ols}}|^\gamma} = \text{sgn}(\hat{\theta}_j) \frac{\lambda_n n^{(0.5\gamma-1)}}{O_p(1)} \rightarrow 0$$

and we cannot rule out the possibility that the two sides of (A.21) being equal.

Finally, to show (22) we argue by contraposition. For those $j \in \hat{\mathcal{C}}$, the counterpart of (A.21) is the following KKT condition

$$x_j^{c'}(y - W\hat{\theta}) = \text{sgn}(\hat{\theta}_j) \lambda_n \hat{\tau}_j / 2, \quad \forall j \in \hat{\mathcal{C}}. \quad (\text{A.23})$$

(21) already rules out $\text{CoRk}(M^*) > \text{CoRk}(\hat{M})$ in the limit. Now suppose $\text{CoRk}(M^*) < \text{CoRk}(\hat{M})$. If so, for any $\hat{M}$ there exists a constant cointegrating vector $\psi \in \mathbb{R}^{p_c}$ such that $\psi'\psi = 1$ (scale normalization), $\psi_{\mathcal{C}^{*c} \cap \hat{\mathcal{C}}} \neq 0$ (must involve wrongly selected inactive variables) and the linear combination $x_i^c \psi$ is an $I(0)$. Although ψ is not unique in general, we use ψ to represent any one of them. Such a ψ linearly combines the multiple equations in (A.23) to generate a single equation

$$\frac{1}{\sqrt{n}} \sum_{j \in \hat{\mathcal{C}}} \psi_j x_j^{c'}(y - W\hat{\theta}) = \frac{\lambda_n}{2\sqrt{n}} \sum_{j \in \hat{\mathcal{C}}} \text{sgn}(\hat{\theta}_j) \hat{\psi}_j \hat{\tau}_j, \quad (\text{A.24})$$

where a scaling factor $n^{-1/2}$ is multiplied on both sides to facilitate the derivation of the rate, and we define $\hat{\psi}_j = \text{sgn}(\hat{\theta}_j)\psi_j$ to simplify notation. The left-hand side of (A.24) is $O_p(1)$ since $\sum_{j \in \hat{C}} \psi_j x_{ji} = O_p(1)$ due to cointegration. The right-hand side of (A.24) can be decomposed into two terms:

$$\frac{\lambda_n}{\sqrt{n}} \sum_{j \in \hat{C}} \text{sgn}(\hat{\theta}_j) \hat{\psi}_j \hat{\tau}_j = \frac{\lambda_n}{\sqrt{n}} \sum_{j \in \hat{C} \cap C^*} \text{sgn}(\hat{\theta}_j) \hat{\psi}_j \hat{\tau}_j + \frac{\lambda_n}{\sqrt{n}} \sum_{j \in \hat{C} \cap C^{*c}} \text{sgn}(\hat{\theta}_j) \hat{\psi}_j \hat{\tau}_j =: I_1 + I_2.$$

The first term

$$I_1 \leq \frac{\lambda_n}{\sqrt{n}} \sum_{j \in \hat{C} \cap C^*} |\hat{\psi}_j| |\hat{\tau}_j| \leq \frac{\lambda_n}{\sqrt{n}} \sum_{j \in \hat{C} \cap C^*} |\hat{\tau}_j| \leq \frac{\lambda_n}{\sqrt{n}} \sum_{j \in C^*} |\hat{\tau}_j| = \frac{\lambda_n}{\sqrt{n}} O_p(1) = o_p(1)$$

as $\hat{\tau}_j = O_p(1)$ for $j \in C^*$, whereas the second term

$$I_2 = \lambda_n n^{0.5(\gamma-1)} \sum_{j \in \hat{C} \cap C^{*c}} \frac{\text{sgn}(\hat{\theta}_j) \hat{\psi}_j}{|\sqrt{n} \hat{\theta}_j^{\text{ols}}|^\gamma} \rightarrow \infty \text{ or } -\infty$$

as $\sqrt{n} \hat{\theta}_j^{\text{ols}} = O_p(1)$ for $j \in C^{*c}$, ψ_j is a constant with nonzero elements, and $\lambda_n n^{0.5(\gamma-1)} \rightarrow \infty$ given the specified rate for $\lambda_n \rightarrow \infty$ and $\gamma \geq 1$. Whether the right-hand side diverges to $+\infty$ or $-\infty$ depends on the configuration of $\hat{\psi}_j$'s and the values of the sign functions. Eq. (A.24) holds with probability approaching 0 when n is sufficiently large. That is, no inactive cointegrating residuals can be formed within the selected variables wpa1. Otherwise, a redundant cointegration group would induce a cointegration vector ψ that sends the right-hand side of (A.24) to either ∞ or $-\infty$ in the limit. ■

Proof of Theorem 4. In Eq. (10) we have separated the cointegrated variables into the active ones and the inactive ones

$$y = Z\alpha + X\beta + u = Z\alpha + \sum_{l \in C^*} X_l^c \phi_l + \sum_{l \in C^{*c}} X_l^c \phi_l + X\beta + u. \quad (\text{A.25})$$

We proceed our argument conditioning on the three events

$$S_1 = \{M^* \cap \mathcal{I} = \hat{M} \cap \mathcal{I}\}, S_2 = \{C^* \subseteq \hat{C}\}, S_3 = \{\text{CoRk}(M^*) = \text{CoRk}(\hat{M})\}.$$

According to Theorem 3, these events occur wpa1, given a sufficiently large sample size.

Under these three events, the regression equation (A.25) is reduced to

$$y = Z_{M^*} \alpha_{M^*} + \sum_{l \in C^*} X_l^c \phi_l + \left(\sum_{l \in C^{*c} \cap \hat{C}} X_l^c \phi_l + X_{M^*} \beta_{M^*} \right) + u, \quad (\text{A.26})$$

where on the right-hand side of the above equation the first and the fourth terms are present by the event S_1 , the second and the third terms by S_2 . Due to event S_3 , there is no cointegration group in the third term, so we can re-write (A.26) as

$$y = Z_{M^*} \alpha_{M^*} + \sum_{l \in C^*} X_l^c \phi_l + \tilde{X}^+ \tilde{\beta}^+ + u, \quad (\text{A.27})$$

where $\tilde{X}^+ = ((X_l^c)_{l \in C^{*c} \cap \hat{C}}, X_{M^*})$ is the collection of augmented non-cointegrating local unit root processes in the post-selection regression equation (A.26) and $\tilde{\beta}^+ = ((\phi_j)_{l \in C^{*c} \cap \hat{C}}, \beta_{M^*})$ is the corresponding coefficient vector.

For $l \in C^{*c} \cap \hat{C}$, the true coefficients are 0. Now these variables appear as non-cointegrating local unit root processes in (A.27), for which Theorem 3 gives variable selection consistency. We implement the post-selection Alasso in (A.27), or equivalently TALasso as in (23). Among the variables in $\tilde{X}$ TALasso will eliminate $(X_l^c)_{l \in C^{*c} \cap \hat{C}}$ wpa1 while the true active variables in M^* will be maintained. In the mean time, all variables in the first and second terms in (A.27) are active and Theorem 3's (21) guarantees their survival wpa1. The asymptotic distribution immediately follows by applying Theorem 3(a) to (A.26). ■

A.3. Proofs in Section 4

In the following proofs concerning Plasso and Slasso, we use a compact notation

$$D(s, v, \theta) = \sum_{j=1}^{\dim(\theta)} s_j [v_j \text{sgn}(\theta_j) I(\theta_j \neq 0) + |v_j| I(\theta_j = 0)]$$

for three generic vectors s , v , and θ of the same dimension. It takes the scalar-based symbol $D(\cdot, \cdot, \cdot)$ in the main text as a special case.

Proof of Corollary 1. For Part (a) and (b), we start with the local perturbation $\check{v} = (\check{v}_z', \check{v}_1', \check{v}_2', \check{v}_x')'$, $v_n = Q^{-1}R_n^{-1}\check{v}$ and $\theta_n = \theta_n^* + v_n$. Notice that $\check{v}$ is different from $\tilde{v}$ in Theorem 3 as we do not impose $\check{v}_2 = 0$. Define

$$V_n(\check{v}) = v_n' W' W v_n - 2v_n' W' u + \lambda_n \sum_{j=1}^p (|\theta_{jn}^* + v_n| - |\theta_{jn}^*|). \quad (\text{A.28})$$

In view of (A.17) and (A.18),

$$V_n(\check{v}) \implies V(\check{v}) = \check{v}' \Omega^+ \check{v} - 2\check{v}' \zeta^+ + \lim_{n \rightarrow \infty} \lambda_n \sum_{j=1}^p R_{jn}^{-1} D(1, \check{v}_j, \theta_j^{0*}).$$

Notice for $j \in \mathcal{I}_1$ with the local-to-zero true coefficient $\theta_{jn}^* = \theta_j^{0*}/n^{\delta_j}$, if $\theta_j^{0*} \neq 0$, then θ_{jn}^* dominates $\check{v}_j/n$ in the limit and $\lambda_n (|\theta_{jn}^* + n^{-1}\check{v}_j| - |\theta_{jn}^*|) = (\lambda_n/n) \text{sgn}(\theta_j^{0*}) \cdot \check{v}_j \rightarrow 0$ for any $\lambda_n/n \rightarrow 0$. Similarly, if $\theta_j^{0*} = 0$, we have $\lambda_n (|\theta_{jn}^* + n^{-1}\check{v}_j| - |\theta_{jn}^*|) = (\lambda_n/n) |\check{v}_j| \rightarrow 0$ when $\check{v}_j \neq 0$. Thus δ_j is irrelevant in the limit for all parts (a), (b) and (c).

We invoke the Convexity Lemma. For Part (a), $\lambda_n/R_{jn} \rightarrow 0$ for all j so the penalty vanishes in the limit and the asymptotic distribution is equivalent to that of OLS. For Part (b), the tuning parameter's rate is $\lambda_n/\sqrt{n} \rightarrow c_\lambda \in (0, \infty)$ so that only the penalty term associated with $\mathcal{I}_1$ vanishes in the limit.

Part (c) needs more elaboration. Let $v_{\lambda_n} = (\lambda_n/\sqrt{n})Q^{-1}R_n^{-1}\check{v} = (\lambda_n/\sqrt{n})v_n$ and $\theta_{\lambda_n} = \theta_n^* + v_{\lambda_n}$. Define

$$V_{\lambda_n}(\check{v}) = v_{\lambda_n}' W' W v_{\lambda_n} - 2v_{\lambda_n}' W' u + \lambda_n \sum_{j=1}^p (|\theta_{jn}^* + v_{\lambda_n,j}| - |\theta_{jn}^*|),$$

and multiply n/λ_n^2 on both sides,

$$\left(\frac{n}{\lambda_n^2}\right) V_{\lambda_n}(\check{v}) = v_n' W' W v_n - 2\frac{\sqrt{n}}{\lambda_n} v_n' W' u + \frac{n}{\lambda_n} \sum_{j=1}^p (|\theta_{jn}^* + v_{\lambda_n,j}| - |\theta_{jn}^*|).$$

By the rate condition of λ_n , the second term $2(\sqrt{n}/\lambda_n)v_n' W' u = o_p(1)$. Given $\check{v}_j \neq 0$ and n large enough:

- If $j \in \mathcal{I}_0 \cup \mathcal{C}$, the coefficients $\theta_{jn}^* = \theta_j^{0*}$ are invariant with n so that

$$\frac{n}{\lambda_n} (|\theta_j^{0*} + \frac{\lambda_n}{\sqrt{n}} \frac{\check{v}_j}{\sqrt{n}}| - |\theta_j^{0*}|) = \frac{n}{\lambda_n} D\left(1, \frac{\lambda_n}{n} \check{v}_j, \theta_j^{0*}\right) = D(1, \check{v}_j, \theta_j^{0*}). \quad (\text{A.29})$$

- If $j \in \mathcal{I}_1$, the reverse triangular inequality $||a+b| - |a|| \leq |b|$ for any $a, b \in \mathbb{R}$ guarantees

$$\frac{n}{\lambda_n} \left| |\theta_{jn}^* + \frac{\lambda_n}{n^{3/2}} \check{v}_j| - |\theta_{jn}^*| \right| \leq \frac{n}{\lambda_n} \left| \frac{\lambda_n}{n^{3/2}} \check{v}_j \right| = O(n^{-1/2}), \quad (\text{A.30})$$

which is dominated by $D(1, \check{v}_j, \theta_j^{0*})$ in the limit if $\check{v}_j \neq 0$ and $\theta_j^{0*} \neq 0$.

Thus we conclude Part (c) ■

Proof of Corollary 2. We first express the limit distributions of the sample standard deviation.

- For a nonstationary variable x_l associated with $\mathcal{I}_1$, we have its limit distribution

$$d_l \sim \sqrt{\int_0^1 J_{c_{x,l}}^2(r) dr - \left(\int_0^1 J_{c_{x,l}}(r) dr\right)^2}.$$

- For $x_{2,l}^c$ associated with $\mathcal{C}_2$, similarly $d_l \sim \sqrt{\int_0^1 J_{c_{2,l}}^2(r) dr - \left(\int_0^1 J_{c_{2,l}}(r) dr\right)^2}$.
- For $x_{1l}^c = \sum_{s=1}^{p_2} A_{1,s} x_{2s}^c + v_{1l}$ associated with $\mathcal{C}_1$, for each i we have $x_{1l}^c = A'_{1,l} x_{2i}^c + v_{1il}$ where $A'_{1,l}$ is the l th row of A_1 matrix, for $l \in (1, \dots, p_1)$. Note that, for each $i = \lfloor nr \rfloor$ with $r \in [0, 1]$,

$$\frac{x_{1il}^c}{\sqrt{n}} = A'_{1,l} \frac{x_{2i}^c}{\sqrt{n}} + \frac{v_{1il}}{\sqrt{n}} \sim A'_{1,l} J_{c_2}(r) + O_p\left(\frac{1}{\sqrt{n}}\right)$$

so that after some simple algebra,

$$\frac{\hat{\sigma}_l}{\sqrt{n}} = \sqrt{\frac{1}{n^2} \sum_{i=1}^n (x_{1il}^c)^2 - \frac{1}{n} \left(\frac{1}{n} \sum_{i=1}^n x_{1il}^c\right)^2} = \sqrt{\frac{1}{n} \sum_{i=1}^n \left(A'_{1,l} \frac{x_{2i}^c}{\sqrt{n}}\right)^2 - \left(\frac{1}{n} \sum_{i=1}^n A'_{1,l} \frac{x_{2i}^c}{\sqrt{n}}\right)^2} + o_p(1)$$

$$\implies d_l \sim \sqrt{A'_{1,l} \left(\int J_{c2}(r) J'_{c2}(r) dr \right) A_{1,l} - \left(A'_{1,l} \int J_{c2}(r) dr \right)^2}.$$

Now we start the proof. Set the same local perturbation $\check{v} = (\check{v}'_z, \check{v}'_1, \check{v}'_2, \check{v}'_x)'$, $v_n = Q^{-1}R_n^{-1}\check{v}$ and $\theta_n = \theta_n^* + v_n$ as in the proof of Corollary 1. We focus on the counterpart of the terms in (A.28).

- If $j \in \mathcal{I}_0$, we have $\hat{\sigma}_j = O_p(1)$ and the coefficient θ_j^{0*} is independent of n so that

$$\hat{\sigma}_j \left(|\theta_j^{0*} + \frac{v_j}{\sqrt{n}}| - |\theta_j^{0*}| \right) = D \left(\hat{\sigma}_j, \frac{v_j}{\sqrt{n}}, \theta_j^{0*} \right) = D \left(O_p(1), O \left(\frac{1}{\sqrt{n}} \right), \theta_j^{0*} \right) \xrightarrow{p} 0;$$

- If $j \in \mathcal{C}$, again θ_j^{0*} is independent of n and

$$\hat{\sigma}_j \left(|\theta_j^{0*} + \frac{v_j}{\sqrt{n}}| - |\theta_j^{0*}| \right) = D \left(\hat{\sigma}_j, \frac{v_j}{\sqrt{n}}, \theta_j^{0*} \right) = D \left(\frac{\hat{\sigma}_j}{\sqrt{n}}, v_j, \theta_j^{0*} \right) \implies D(d_j, v_j, \theta_j^{0*}) = O_p(1),$$

since these indices are associated with a local unit root process x_j^c and therefore $\hat{\sigma}_j/\sqrt{n} \implies d_j$ is non-degenerate;

- If $j \in \mathcal{I}_1$, similarly we have

$$\hat{\sigma}_j \left(|\theta_{jn}^* + \frac{v_j}{n}| - |\theta_{jn}^*| \right) = D \left(\hat{\sigma}_j, \frac{v_j}{n}, \theta_j^{0*} \right) = D \left(\frac{\hat{\sigma}_j}{\sqrt{n}}, \frac{v_j}{\sqrt{n}}, \theta_j^{0*} \right) = D \left(O_p(1), \frac{v_j}{\sqrt{n}}, \theta_j^{0*} \right) \xrightarrow{p} 0.$$

The above analysis implies

$$V_n(\check{v}) \implies V(\check{v}) = \check{v}' \Omega^+ \check{v} - 2\check{v}' \zeta^+ + c_\lambda \sum_{j \in \mathcal{C}} D(d_j, v_j, \theta_j^{0*}), \quad (\text{A.31})$$

and Part (a) and (b) follow.

For Part (c), let $R_n = R_n/\lambda_n$ and $\theta_n = \theta_n^* + \tilde{R}_n^{-1}\check{v}$. Define

$$\tilde{V}_n(\check{v}) = \check{v}' \left(\tilde{R}_n^{-1} W' W \tilde{R}_n^{-1} \right) \check{v} - 2\check{v}' \tilde{R}_n^{-1} W' u + \lambda_n \sum_{j=1}^p \hat{\sigma}_j (|\theta_{jn}^* + \tilde{R}_{jn}^{-1} \check{v}_j| - |\theta_{jn}^*|).$$

Multiply $1/\lambda_n^2$ on both sides,

$$\begin{aligned} \frac{\tilde{V}_n(\check{v})}{\lambda_n^2} &= \check{v}' (R_n^{-1} W' W R_n^{-1}) \check{v} - \frac{2}{\lambda_n} \check{v}' R_n W' u + \frac{1}{\lambda_n} \sum_{j=1}^p \hat{\sigma}_j (|\theta_{jn}^* + \tilde{R}_{jn}^{-1} \check{v}_j| - |\theta_{jn}^*|) \\ &= \check{v}' (R_n^{-1} W' W R_n^{-1}) \check{v} + o_p(1) + \frac{1}{\lambda_n} \sum_{j=1}^p \hat{\sigma}_j (|\theta_{jn}^* + \tilde{R}_{jn}^{-1} \check{v}_j| - |\theta_{jn}^*|). \end{aligned}$$

by the rate condition of λ_n . Again we study the last term. For $\check{v}_j \neq 0$ and a sufficiently large n :

- For $j \in \mathcal{I}_0$,

$$\frac{1}{\lambda_n} \hat{\sigma}_j \left(|\theta_j^{0*} + \frac{\lambda_n}{\sqrt{n}} \check{v}_j| - |\theta_j^{0*}| \right) = \frac{1}{\lambda_n} D \left(\hat{\sigma}_j, \frac{\lambda_n}{\sqrt{n}} \check{v}_j, \theta_j^{0*} \right) = D \left(\hat{\sigma}_j, \frac{\check{v}_j}{\sqrt{n}}, \theta_j^{0*} \right) \xrightarrow{p} 0;$$

- For $j \in \mathcal{C}$,

$$\begin{aligned} \frac{1}{\lambda_n} \hat{\sigma}_j \left(|\theta_j^{0*} + \frac{\lambda_n}{\sqrt{n}} \check{v}_j| - |\theta_j^{0*}| \right) &= \frac{1}{\lambda_n} D \left(\hat{\sigma}_j, \frac{\lambda_n}{\sqrt{n}} \check{v}_j, \theta_j^{0*} \right) \\ &= D \left(\frac{\hat{\sigma}_j}{\sqrt{n}}, \check{v}_j, \theta_j^{0*} \right) = D(d_j, v_j, \theta_j^{0*}) = O_p(1); \end{aligned}$$

- For $j \in \mathcal{I}_1$, the reverse triangular inequality $||a+b| - |a|| \leq |b|$ implies

$$\frac{1}{\lambda_n} \hat{\sigma}_j \left| |\theta_{jn}^* + \frac{\lambda_n}{n} \check{v}_j| - |\theta_{jn}^*| \right| \leq \frac{1}{\lambda_n} \hat{\sigma}_j \frac{\lambda_n}{n} \check{v}_j = \frac{\hat{\sigma}_j}{\sqrt{n}} \frac{\check{v}_j}{\sqrt{n}} \xrightarrow{p} 0.$$

We obtain $\lambda_n^{-2} \tilde{V}(\check{v}) \implies \check{v}' \Omega^+ \check{v} + \sum_{j \in \mathcal{C}} D(d_j, v_j, \theta_j^{0*})$ and the conclusion follows. ■

Appendix B. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.jeconom.2021.02.002>.

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